Thermodynamic

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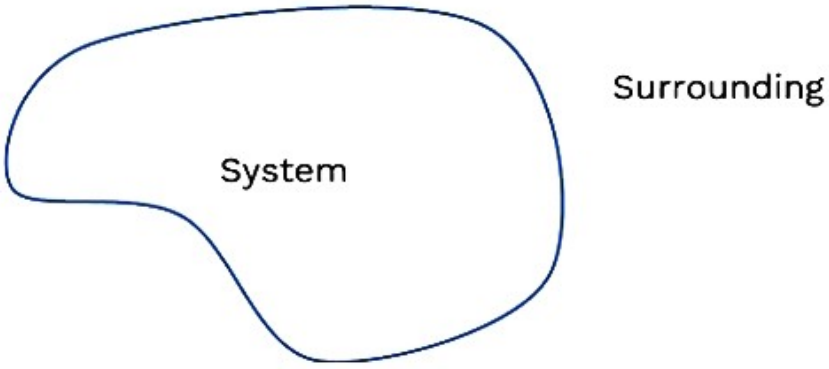
Thermodynamics

1. It is the study of energy & its exchange during various physical & chemical processes.
2. It also gives information about the spontaneity of a process under certain conditions theoretically.

Basic terminology of thermodynamics-

System & Surroundings-

1. The part of the universe under thermodynamic observation is known as the system.
2. The rest part of universe is called surroundings
   i.e. System + Surrounding = Universe
3. In thermodynamic observation, generally size of the surrounding is very-very large as compared to the system.
4. Boundary/wall - The interface which separates the system from its surroundings knows as Boundary/Wall.



5. Any exchange between system & surroundings takes place through the boundary.
6. Boundary may be real ↓ can not exchange matter or imaginary ↓ can exchange matter
7. Boundary may be adiabatic ↓ can not exchange heat or diathermic ↓ can exchange heat
8. Boundary may be movable ↓ can exchange work or immovable ↓ can not exchange work

Type of system

On the basis of the ability to exchange matter & energy. There are three types of systems as following:

i) Open system:

A system that can exchange both matter & energy with the surroundings is known as an open system.

At least one wall must be imaginary.

e.g. Jug, flask cup, mug etc.

Points to remember

Between system & surroundings. Energy can be exchanged as heat or work only.

ii) Closed system:

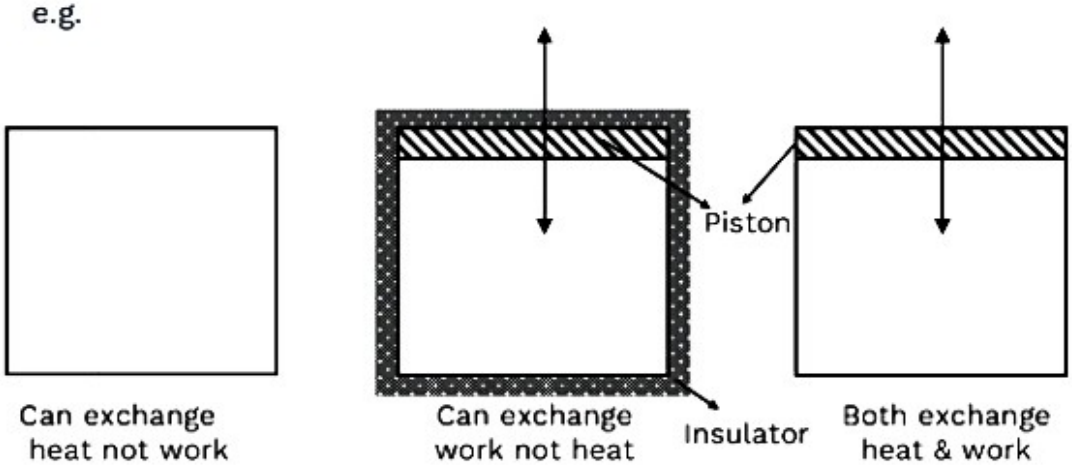
A system that can exchange energy but not matter with the surroundings is known as a closed system.

All walls must be real.

e.g.

Points to remember

Earth is an open system.



iii) Isolated system:

A system that can not exchange matter & energy (heat and work) both with the surroundings is known as an isolated system.

This means all walls must be real adiabatic & immovable.

e.g.



Thermodynamic equilibrium:

It consists of following three parts –

(i) Material Equilibrium (amount is constant)

(ii) thermal Equilibrium (Temperature is constant)

(iii) Mechanical Equilibrium (Pressure & volume constant)

When all three equilibrium is simultaneously established, then the system is said to be present in thermodynamic equilibrium or in a state.

State Functions/State variables-

Measurable properties which can be defined at a particular state of the system are known as a state function.

e.g.

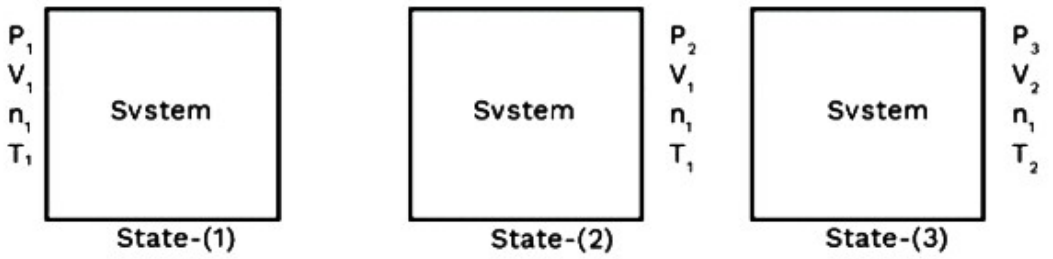
Mass, Moles, Temperature, volume, pressure, etc.

At a particular state, all stable functions have specific fixed values.

Two different forms of the same system have at least one different value of state function.

Points to remember

Thermodynamic universe is an example of an isolated system.



Between two states, the change in state function is the same for all paths & depends upon initial & final states only.

Let x is a state function that is changing:

Total change in x

For 1st path = $\Delta X_1 + \Delta X_2$

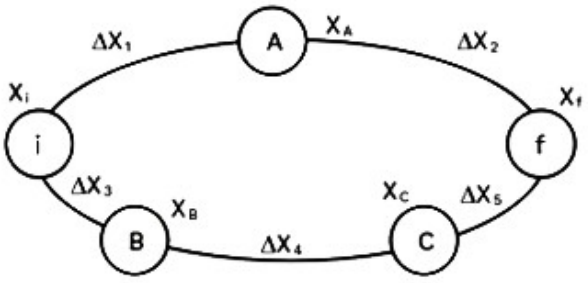
$= (X_A - X_i) + (X_f - X_A) = (X_f - X_i)$

For 2nd path = $\Delta X_3 + \Delta X_4 + \Delta X_5$

$= (X_B - X_i) + (X_C - X_B) + (X_f - X_C) = (X_f - X_i)$

i.e. for ∞ step process-

dx – infinitesimally small change in X



So, Total change in $X = \int_i^f dx$

$= X_f - X_i$

$= \Delta X$

Path functions

Quantities which are defined at the path, as well as their values, depending upon the path followed by the system.

e.g.

heat (q) & work (w)

Let Y is a path function -

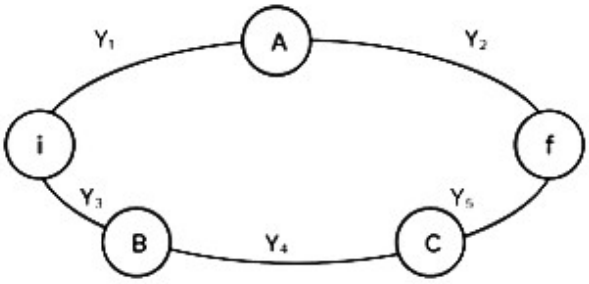
Total value of Y,

For 1st path = $Y_1 + Y_2$

For 2nd path = $Y_3 + Y_4 + Y_5$

So, For infinite step process,

dY – very small value of Y



$\Rightarrow$ Total value of $Y = \int_i^f dY$

$= Y$

i.e., $\int_i^f dT = \Delta T$

$\int_i^f dV = \Delta T$

But $\int_i^f dq \neq \Delta q$

$= q$

$\int_i^f dw \neq \Delta W$

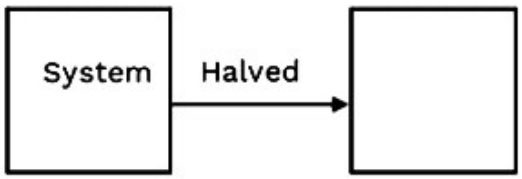
Types of measurable properties

Extensive properties	Intensive properties
Properties that depend upon the amount of system are known as extensive properties. e.g., Mass, moles, volume, Heat capacity Internal energy (U), Enthalpy (H) Entropy (s), Gibbs free energy (G) Resistance etc.	Properties that do not depend upon the amount of system are known as intensive properties. e.g., Temperature, pressure, density, molar mass, molarity, molality, melting point, boiling point, molar U, molar H, Molar G, pH of the solution, electrode potential, etc.

Mass=w
Volume = V
Temperature = T

$\text{Density(d)} = \frac{w}{v}$

$\text{Molarity(M)} = \frac{n}{v}$



Mass=w/2 ⇒ extensive
Volume = V/2 ⇒ extensive
Temperature = T⇒ intensive

$\text{Density} = \frac{w/2}{v/2} = (d) \Rightarrow$ intensive

$\text{Molarity} = \frac{n/2}{v/2} = (M) \Rightarrow$ intensive

Process-

The phenomenon obeying which the system moves from one state to another state.
From a particular state, infinite types of the process can be started.
Between two fixed states, there can be one process that can be done in two ways as reversible or irreversibly.
If the system & surroundings can be restored back to their original state just by reversing the direction of the driving force, then the process is reversible, otherwise irreversible.

Points to remember

$\frac{\text{extensive}}{\text{extensive}} = \text{intensive}$

Reversible & Irreversible process :

Irreversible process	Reversible process
(i) It is a finite step process.	(i) It is an infinite step process.
(ii) It is a finite difference between two successive states.	(ii) It is an infinitesimally slight difference between two successive states.

Irreversible process	Reversible process
(iii) It is a fast process.	(iii) It is a very slow process.
(iv) It is a real process which can be achieved practically.	(iv) It is a hypothetical process means can be achieved only theoretically.
(v) In this process only initial & final states are known; hence cannot be plotted on a graph.	(v) In this process initial & final states, as well as the whole path, are known; hence can be plotted on a graph.

Points to remember

Reversible reaction must not be considered as a thermodynamically reversible process. In fact, reversible reactions are actually irreversible processes. (actually all reactions)

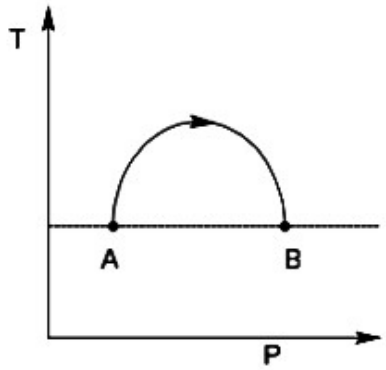
Isothermal Process:

The process in which the temperature of the system remains constant throughout the process is known as the isothermal process.

$T \longrightarrow$ constant

$\Rightarrow dT = 0$

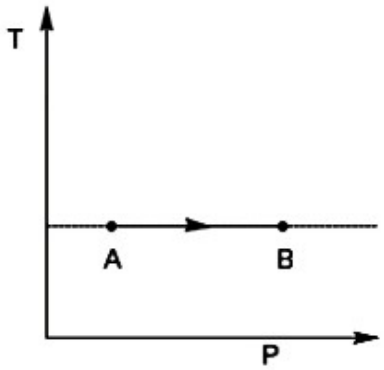
$\Rightarrow \Delta T = 0$



$\Delta T = 0$

but not isothermal

because $dT \neq 0$



isothermal

i.e. $dT = 0$

$\Rightarrow \Delta T = \int dT = 0$

Isobaric process:

The process in which the pressure of the system remains constant throughout the process is known as the isobaric process.

$P \longrightarrow$ constant

$\Rightarrow \boxed{dP = 0}$

Isochoric process:

The procedure in which the volume of the system remains constant throughout the process is known as the isochoric process.

Volume $\longrightarrow$ Constant

$\Rightarrow \boxed{dv = 0}$

Adiabatic process:

The process in which no heat exchange throughout the process, known as the adiabatic process.

i.e. $\Rightarrow \boxed{dq = 0}$

$\downarrow$

very small heat exchange

Cyclic process:

If the system Undergoes a series of the process but finally comes back to the original state, the whole process is known as a cyclic process.

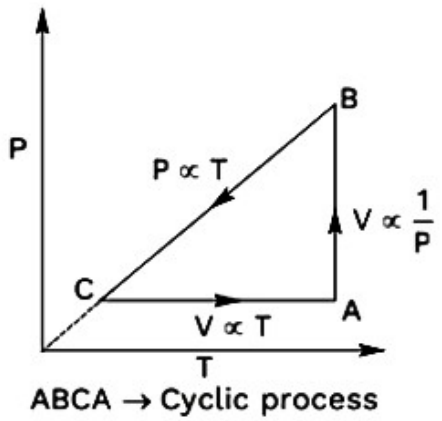
Means in a cyclic process initial & final steps are always same; hence the net change in any state function for the overall cyclic process is zero.

Let X is a state function

$$\boxed{\oint dx = 0}$$

Example:

(i) For ideal gas (n mole)-



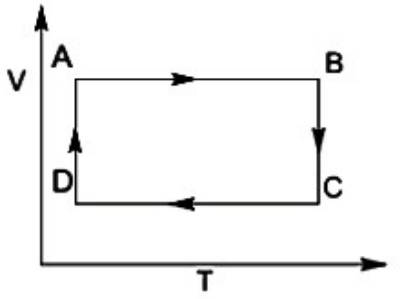
ABCA → Cyclic process

AB $\longrightarrow$ Reversible isothermal compression

BC $\longrightarrow$ Reversible isochoric cooling

CA $\longrightarrow$ Reversible isobaric heating or expansion.

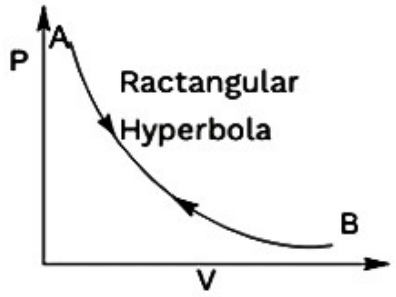
(ii) For ideal gas-



ABCDA → Cyclic process

AB ⟶ Isochoric heating
BC ⟶ Isothermal compression
CD ⟶ Isochoric cooling
DA ⟶ Isothermal expansion

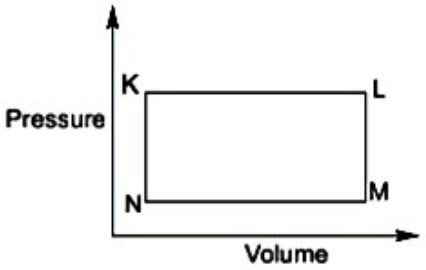
(iii) For ideal gas



ABA → Cyclic process

ABA ⟶ Cyclic process
AB ⟶ Isothermal expansion
BA ⟶ Isothermal compression

Q1 **A fixed mass 'm' of gas is subjected to the transformation of states from K to L to M to N and back to K, as shown in the figure. The pair of isochoric processes among the transformation of states is:**
(A) K to L and L to M
(B) L to M and N to K
(C) L to M and M to N
(D) M to N and N to K

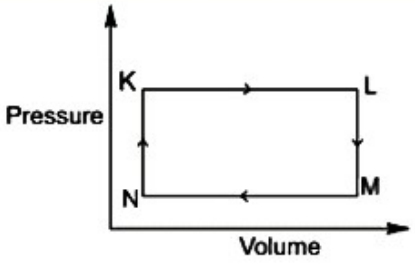


Sol. **(B)**

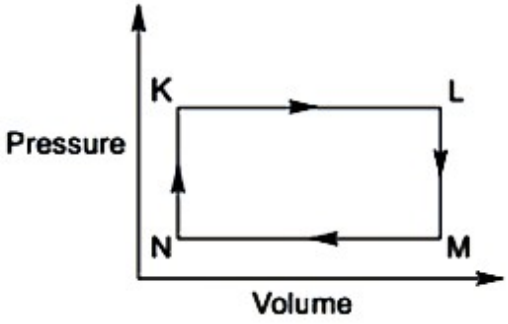
In case of C to M and N to K

Volume is constant so this process

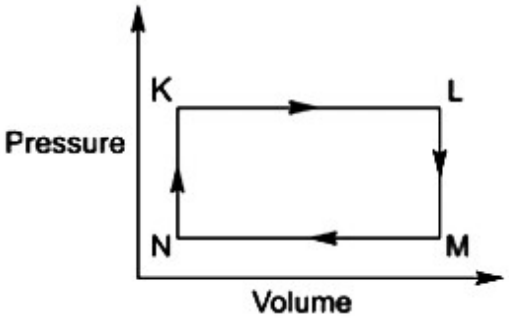
are known as isochoric process.



Q2 **A fixed mass 'm' of gas is subjected to the transformation of states from K to L to M to N and back to K, as shown in the figure.**
The succeeding operations that enable this transformation of states are:
(A) Heating, cooling, heating cooling
(B) Cooling, heating, cooling, heating
(C) Heating, cooling, cooling, heating
(D) Cooling, heating, heating, cooling



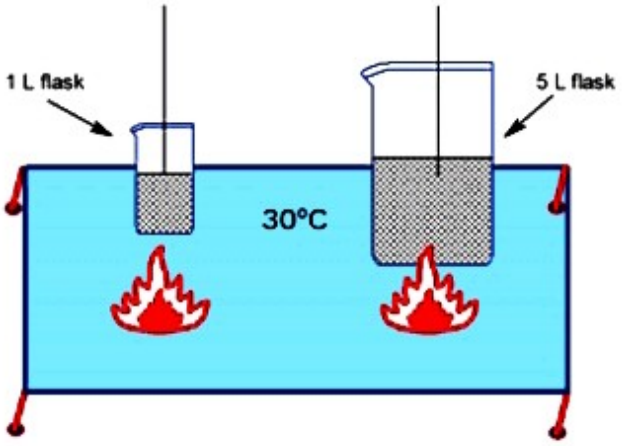
Sol: (C)
K to L $\Rightarrow V \propto T \Rightarrow$ Heating (Because V ↑)
K to M $\Rightarrow P \propto T \Rightarrow$ Cooling (Because P ↓)
M to N $\Rightarrow V \propto T \Rightarrow$ Cooling (Because V ↓)
N to K $\Rightarrow P \propto T \Rightarrow$ Heating (Because P ↑)



Heat capacity (C):
The quantity of heat absorbed/ released by a substance to increase/ decrease its temperature by 1°C (or by 1 Kelvin)
For a substance,

$$\boxed{C = \frac{dq}{dT}}$$ dq = Heat, dT = change in temperature

Units-
J/K = J/°C
Cal = Cal/°C
It is an extensive property.



Specific Heat capacity (C_s):

The amount of heat required/released by one gram of a substance to change its temperature by 1°C.

Means it is heat capacity of one gram substance.

i.e. $C_s = \frac{C}{M}$

$\Rightarrow \boxed{C_s = \frac{dq}{mdT}}$ J / gm k $\frac{Cal}{gm.k}$

It is an intensive property.

Points to remember

For a substance heat capacity are generally measured at two special conditions:

i) At constant volume: C_v, $C_{v,s}$, $C_{v,m}$

ii) At constant pressure: C_p, $C_{p,s}$, $C_{P,M}$

iii) Neither P nor V constant: C, C_s, C_M

Molar Heat Capacity (C_m):-

It is heat capacity of one mole substance

i.e., $C_m = \frac{C}{n}$

$\because$ n = mole of substance

$\Rightarrow \boxed{C_m = \frac{dq}{ndT}}$ J/mol. K or Cal/mol. K

It is an Intensive property.

i.e. A substance have 3 types of molar heat capacity-

If pressure is constant $\Rightarrow C_{p,m}$

If volume is constant $\Rightarrow C_{v,m}$

Otherwise $\Rightarrow C_M$

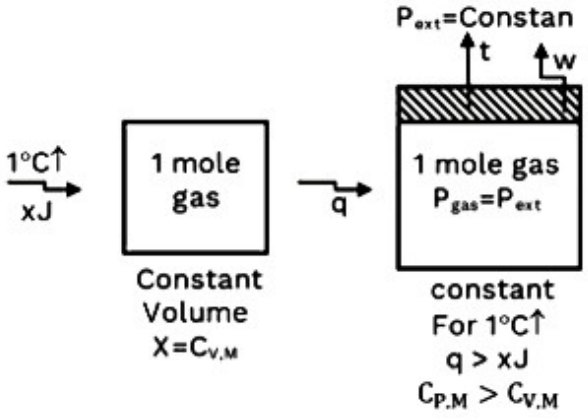
For solids and liquids

$C_m \simeq C_{p,m} \simeq C_{v,m}$

But for gases

$C_m \neq C_{p,m} \neq C_{v,m}$

Always $\boxed{C_{P,M} > C_{V,M}}$



This is because of constant pressure; part of the heat is utilized in work.

For ideal gases:

$\boxed{C_{P,M} - C_{V,M} = R}$ (Gas constant = 8.314 J/mol.

or R = 2 Cal/mol K)

$\boxed{\frac{C_{P,M}}{C_{V,M}} = \gamma}$ Poison's ratio or adiabatic complete

$\Rightarrow \boxed{C_{V,M} = \frac{R}{\gamma - 1}}$

Points to remember

For a substance, heat capacity may be temperature independent (by default) or maybe temperature-dependent (If given in question)

Determination of $C_{V,M}$ for an ideal gas:

Particles of the gas may have three types of degree of freedom as follows:

i) Translational of the degree of freedom (T_f)

It is due to the linear motion of the particle.

For any gas, $\boxed{T_f = 3}$ always.

ii) Rotational degree of freedom (R_f)

Due to the rotational motion of the particle.

- Only polyatomic particles show rotational motion.
- No. of rotational degree of freedom depends upon the shape of the particle.

$$\boxed{\begin{array}{l}\text{For linear particle } R_F = 2\\ \text{For non-linear particle } R_F = 3\end{array}}$$

iii) Vibrational degree of freedom (V_f)

Due to vibrational motion, which occurs only at high temperatures.

$$\boxed{\begin{array}{l}\text{If inactive (by default); } V_F = 0\\ \text{If active (If given in question) } V_F = (3N - T_F - R_F)\end{array}}$$

N→ atomicity of the particle

For an ideal gas:

$$C_{V,M} = T_F\left(\frac{R}{2}\right) + R_F\left(\frac{R}{2}\right) + V_F(R)$$

R → Gas constant

e.g. for monoatomic gas

$$C_{V,M} = 3\left(\frac{R}{2}\right) + 0\left(\frac{R}{2}\right) + 0(R)$$

$$\boxed{\begin{array}{l}C_{v,m} = \dfrac{3R}{2}\\ C_{p,m} = \dfrac{5R}{2}\\ \gamma = \dfrac{5}{3}\end{array}} \rightarrow C_{p,m} - C_{v,m} = R$$

Points to remember

Particles of monoatomic gases can never show rotational & vibrational motions; hence their heat capacities are always temperature independent.

Q3 **Determine $C_{V,M}$ for N_2 and $C_{P,M}$ for CH_4 gas**

Sol

For N_2

$$C_{V,M} = 3\left(\frac{R}{2}\right) + 2\left(\frac{R}{2}\right) + 0$$

$$= \frac{5R}{2}$$

For CH_4

$$C_{V,M} = 3\left(\frac{R}{2}\right) + 3\left(\frac{R}{2}\right) + 0$$

$$= 3R$$

$$C_{P,M} = 4R$$

Q4 **Determine Y for SO_3 gas if vibrational mode is active**

Sol

$$C_{V,M} = 3\left(\frac{R}{2}\right) + 3\left(\frac{R}{2}\right) + (12 - 3 - 3)(R) = 9R$$

$$C_{P,M} = 10\ R$$

$$\Rightarrow \quad \gamma = \frac{10}{9}$$

Internal energy (U):

Internal energy is a state function & an extensive property.

For a system, U is the summation of all types of molecular energies as following-

i.e.,

$$U = \underset{\text{Kinetic Energy}}{\text{Translational}} + \underset{\text{Kinetic Energy}}{\text{Rotational}} + \underset{\text{Kinetic Energy}}{\text{Vibrational}} + \text{Nuclear energy} + \text{electronic energy} \ldots\ldots\ldots\ldots$$

In simple words,

$$U_{system} = \sum \underset{\downarrow}{(K.E.)_{Particles}} + \sum \underset{\downarrow}{(P.E.)_{Particles}}$$

So, In general,

$$\boxed{U = f(T.V)}$$

For an ideal gas,

$\therefore$ No attraction or repulsion among particles.

$\Rightarrow (PE)_{Particles} = 0$

i.e. $\boxed{(U)_{ideal} = f(T) \text{ only}}$

The absolute value of U at a particular state can not be calculated. We can only determine the change in U (ΔU) between two states.

Total energy (E):

For system,

$T.E. = (K.E.)_{system} + (P.E.)_{system} + (I.E.)_{system}$

In thermodynamic observations, the system is considered at rest (K.E.=0) and in the absence of external fieled (P.E.=0) means for a thermodynamic system,

$TE(E) = IE(U)$

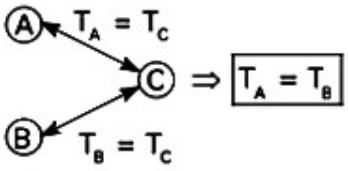
$\Rightarrow \Delta E = \Delta U$

Laws of Thermodynamics:

These laws are applicable only & only for the macroscopic amount of the system, not for a microscopic number of particles.

Zeroth law:

If two systems are in thermal equilibrium with a third system separately, then they both must be in thermal equilibrium with each other too.



The first law of thermodynamics:

It is related to the conservation of energy.

In every thermodynamic process, the total energy of the universe remains conserved.

Consider a thermodynamic system that is subjected to a process exchanging both heat & work. So, the change in U can be given as follows-

$$\boxed{dU = dq + dw}$$

dq → Heat exchanged

dw → work done

dU → change in I.E.

For isolated system (universe),

No energy exchange $\Rightarrow dq = 0$

$dw = 0$

so,

$dU = 0 + 0$

$\Rightarrow dU = 0$

$\Rightarrow dE = 0$

i.e. E is conserved

Now, for the overall process,

$$\int_i^f du = \int_i^f dq + \int_i^f dw$$

$$\boxed{\Delta U = q + w}$$

Where q – Total heat exchanged
w – total work done
ΔU – Change in internal energy
} for overall process

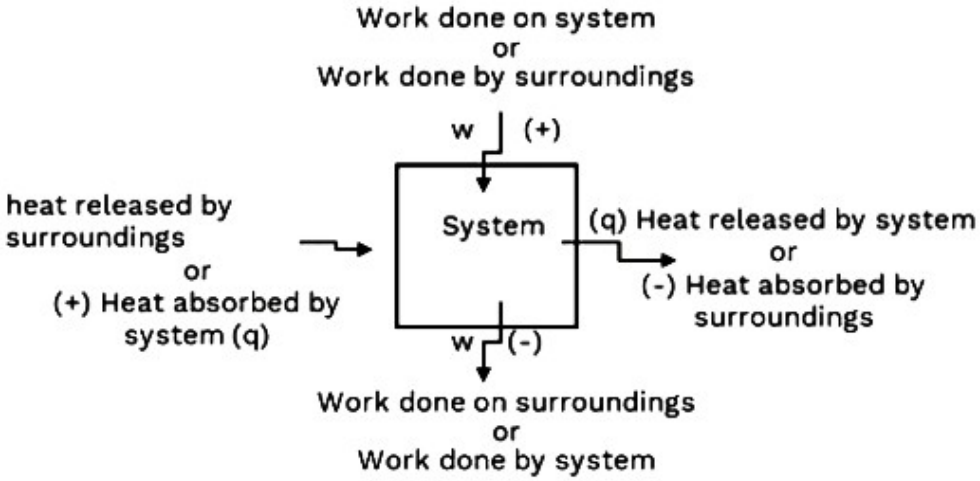
Note: In the above relation, put all values in the same units & with the proper sign.

1 cal = 4.18 J = 4.2 J

1 L.atm = 101.3 J

1 L.bar = 100 J

IUPAC sign convention:



Also

If IE ↑ ⇒ ΔU = (+)

IE ↓ ⇒ ΔU = (–)

Q5 **When 100 J of work is done on the gas, its internal energy decreases by 10 cal. Calculate heat exchanged in the process.**

Sol: $\Delta U = q + w$

$(-10 \text{ cal.}) = q + (+100 \text{ J})$

$q = (-10 \times 4.2\text{J}) - 100 \text{ J}$

$= -142 \text{ J}$

i.e. heat released = 142 J

Q6 **During compression of a spring, the work done is 10 kJ, and 2 kJ escaped to the surroundings as heat. The change in internal energy ΔU (in kJ) is:**
(A) 12 **(B) –8** **(C) 8** **(D) –12**

Sol: (C)

$w = +10$ kJ

$q = -2$ kJ

$\Delta U = q + w$

$= 8$ kJ

Work (w):

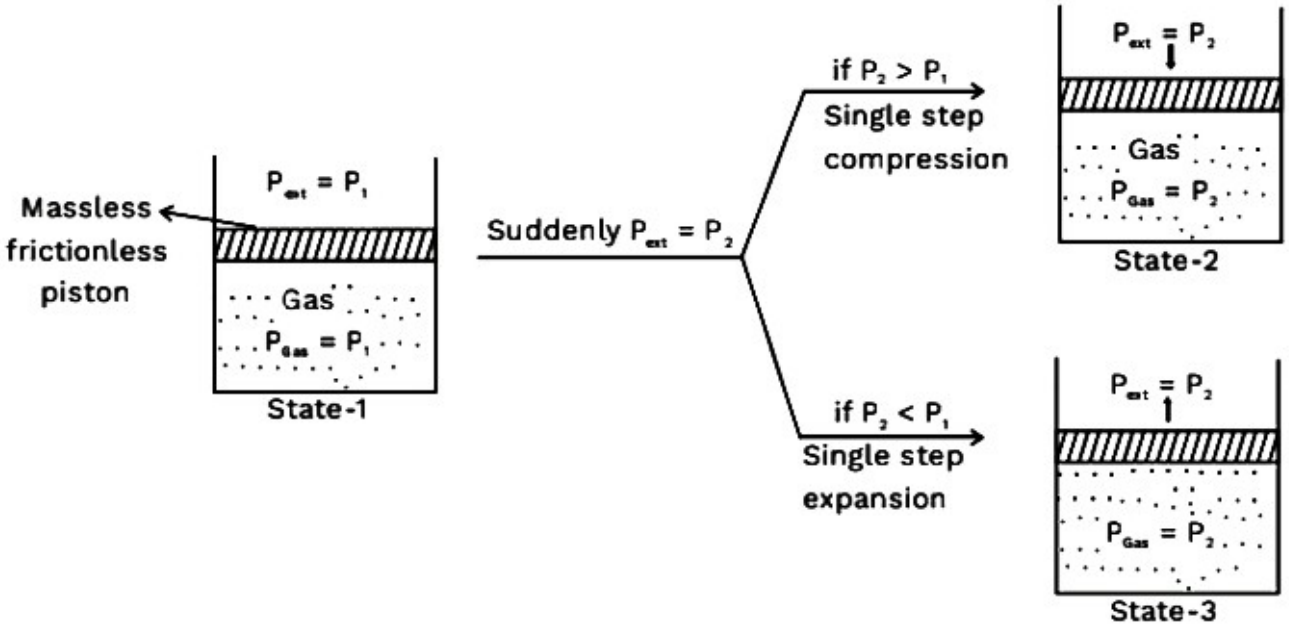
It is an ordered form (or less random form) of energy exchange between system & surrounding.

There are two types of work:

(i) Non–PV work,
Works other than expansion/compression.
e.g., Electrical work
e.g., Gravitational work
e.g., Muscular work etc.

(ii) PV work,
It involves volume change due to pressure
i.e., expansion & compression

For solids & liquids, PV work is negligible means it is significant for gases only.



⇒ So, in both expansion & compression, the work is done due to external pressure P_2 (final pressure of gas), Which can be given as following

For very very small change

$$= \boxed{dw = -P_{ext}.dv}$$

If expansion $\Rightarrow V \uparrow \Rightarrow dv > 0 \Rightarrow dw < 0$

$\Rightarrow$ work done by system

If compression $\Rightarrow V \downarrow \Rightarrow dv < 0 \Rightarrow dw < 0$

$\Rightarrow$ work done on system

⇒ Since work is a path function, so its value depends upon the type of path,

If irreversible path:

$P_{ext} \rightarrow$ constant

$$W_{irr} = \int_i^f dw = -P_{ext} \cdot \int dv$$

If reversible path $(\infty - \text{step})$

$P_{ext} = (P_{gas} \pm dP)$ at every moment

So, $dw_{rev} = -(P_{gas} \pm dP).dv$

$= -P_{gas}.dv \pm \cancel{dp.dv}$ (Negligible)

$= -P_{gas}.dv$

Work calculation for ideal gas

(1) **In isobaric process:** only process for which both reversible and irreversible Paths have the same value of work.

$\because \quad dw = -P_{ext}.dv$

Points to remember

$$= \boxed{dw_{irr} = -P_{ext}.dv}$$

$\downarrow$

constant

$$= \boxed{dw_{irr} = -P_{gas}.dv}$$

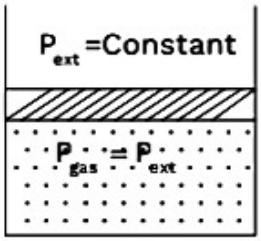
$\downarrow$

varible

Points to remember

If sudden / fast / finite-step, against constant P_{ext} etc. = Irreversible

If slow/gradually/ ∞ – step/graphical etc. = Reversible



$$= -P_{gas}.dv$$

↓

Constant

$$\Rightarrow \boxed{w = -P_{gas}.\Delta V}$$

$\because PV = nRT$

$\Rightarrow P.dv = n.R.dT$

$$\Rightarrow \boxed{W = -n.R.\Delta T}$$

$n \rightarrow$ moles of ideal gas

$R = 8.314 J/mol.k = 2 cal/mol.k$

If heating $\Delta T > 0 \Rightarrow W < 0 \Rightarrow$ expansion

If cooling $\Delta T < 0 \Rightarrow W > 0 \Rightarrow$ compression

(2) Isothermal ideal gas expansion/compression : (dT = O)

(i) If process is Irreversible:

$$\because dw_{irr} = -P_{ext}.dv$$

↓

(Constant)

$$\boxed{W_{irr} = \begin{cases} -P_{ext}.\Delta V \\ -P_{ext}.(V_2 - V_1) \\ -P_{ext}.\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right) \end{cases} P_2 = P_{ext}}$$

(ii) If the process is reversible:

$$\because dW_{rev} = -P_{gas}.dV$$

$$= \frac{-nRT}{V} dV$$

$$\int_i^f dw = -nRT \int_{V_1}^{V_2} \frac{1}{V}\, dv$$

Ideal isothermal $P_1V_1 = P_2V_2$

$$\boxed{W_{rev\,iso} = -nRT \ln \frac{V_2}{V_1} = -nRT \ln \frac{P_1}{P_2}}$$

(3) For chemical reactions & phase transitions :

→ Always take place at a constant temperature.

(i) If volume is constant

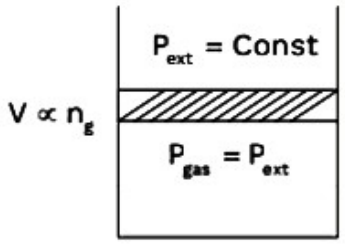
i.e., reaction in closed rigid vessel

$dv = 0$

$\Rightarrow \boxed{w = 0}$

(ii) If pressure is constant

i.e., open vessel or piston fitted vessel



$dw = -P_{ext}.dv$

$= -P_{gas}.dv$

$\boxed{w = -P_{ext}.\Delta V}$

$PV = n_gRT$

$P.dV = dn_gRT$

$dw = -dn_g.R.T$

$\Rightarrow \boxed{w = -\Delta n_g.R.T}$

e.g., $N_2O_4(g) \rightarrow 2NO_2(g)$

$\Delta n_g = (2) - (1) = 1$

> 0

i.e., $n_g \uparrow \Rightarrow w < o$

Note: Above relations is applicable only when at least one ideal gas is present.

e.g., $X(s) \xrightarrow{Patm} X(l)$

No ideal gas

$W = -\Delta n_gRT$ not applicable

But $W = -P_{ext}.\Delta V$

$= -P\left(V_{x(l)} - V_{x(s)}\right)$

Q7 **In which of the following changes work is done on the system, if all changes occur at constant pressure.**

(A) $2O_3(g) \rightarrow 3O_2(g)$

(B) $H_2(g) + I_2(g) \rightarrow 2HI(g)$

(C) $MgSO_4.7H_2O(s) \rightarrow MgSO_4(s) + 7H_2O(g)$

(D) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Sol: **D**

(A) $2O_3(g) \rightarrow 3O_2(g)\ \Delta n_g = 1$

(B) $H_2(g) + I_2(g) \rightarrow 2HI(g)\ \Delta n_g = 0$

(C) $MgSO_4.7H_2O(s) \rightarrow MgSO_4(s) + 7H_2O(g)\ \Delta n_g = 7$

(D) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)\ \Delta n_g = (-2)$

$w = (+)ve$ for work done on the system

$w = -\Delta n_g RT$

$\Rightarrow \Delta n_g < 0$

Q8 **Calculate the magnitude of work done (in kJ) during vaporization of 78 gm benzene (liquid) at its boiling point against 1 atm external pressure. Boiling point of benzene (liquid) = 360 K.[Take : R = 25/3 J/K.mol]**

Sol:

$$\underset{1mole}{C_6H_6(l)} \longrightarrow \underset{1mole}{C_6H_6(g)}$$

$w = -\Delta ngRT$

$= -(1) \times \frac{25}{3} \times 360$

$= -3000J$

$= -3kJ$

Q9 **An ideal gas expands in volume from $1\times10^{-3}m^3$ to $1\times10^{-2}m^3$ at 300 K against a constant pressure of $1\times10^{5}Nm^{-2}$. The work done is-**

(A) - 900J **(B) - 900 kJ** **(C) 2710 kJ** **(D) 900 kJ**

Sol: **A**

$$dw_{irr} = -P_{ext}.dV$$

$$w_{irr} = -P_{ext}.(V_2 - V_1)$$

$$= -1\times10^{5}\left(1\times10^{-2} - 1\times10^{-3}\right)J$$

$$= -10^{5}\times9\times10^{-3}J$$

$$= -900\,J$$

Q10 **A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37°C. As it does so, it absorbs 208 J of heat. The values of w and q for the procedure will be: (R = 8.314 J / mol K) (ln 7.5 = 2.01)**

(A) q = -208 J, w = -208 J **(B) q = -208 J, w = +208 J**

(C) q = +208 J, w = +208 J **(D) q = +208 J, w = -208 J**

Sol: **D**

(R = 8.314 J / mol K) (ln 7.5 = 2.01)

$$\boxed{q = +208\,J}$$

$$dw_{rev} = -\frac{nRT}{V}.dV$$

$$w = -nRT\ln\frac{V_2}{V_1}$$

$$= -0.04\times8.314\times310\ \ln\frac{375}{50}$$

$$= -100.09\ln(7.5) \Rightarrow -103.09\times2.01 \Rightarrow -207.21 \cong -208J$$

Work done in graphical problems (P v/s V graph)

$\because$ graphical $\Rightarrow$ Reversible process

$$\Rightarrow w = -\int_{V_1}^{V_2} P_{gas}.dv$$

i.e., $\boxed{|w| = \text{Area under the curve}}$

also, if volume $\uparrow \Rightarrow w < 0$

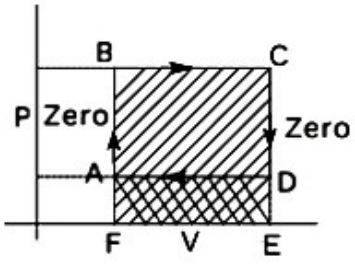
if volume $\downarrow \Rightarrow w > 0$

Work done in cyclic process

$\boxed{w_{total} = w_1 + w_2 + w_3 + ...}$ all with sign

If P vs V graph is given:

$\boxed{|w_{total}| = \text{Area enclosed in the cycle}}$



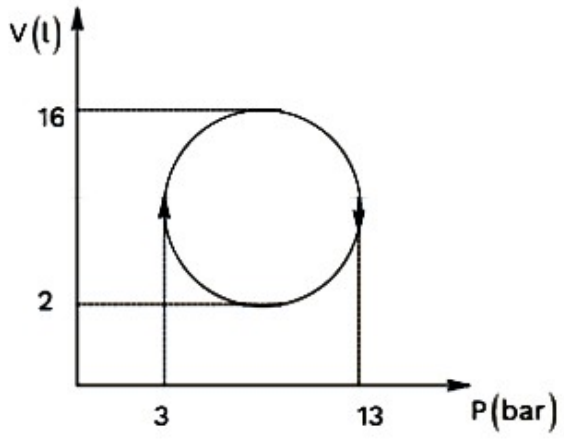
Also: If clockwise $\Rightarrow w = (-)$

If anti clockwise $\Rightarrow w = (+)$

$w_{B \to C} = -(BCDA)$

$w_{D \to A} = -(DAFE)$

Q11 **Work done (in kJ) by the gas in the following cyclic process is**



(A) −11 **(B) −11000** **(C) 11** **(D) 11000**

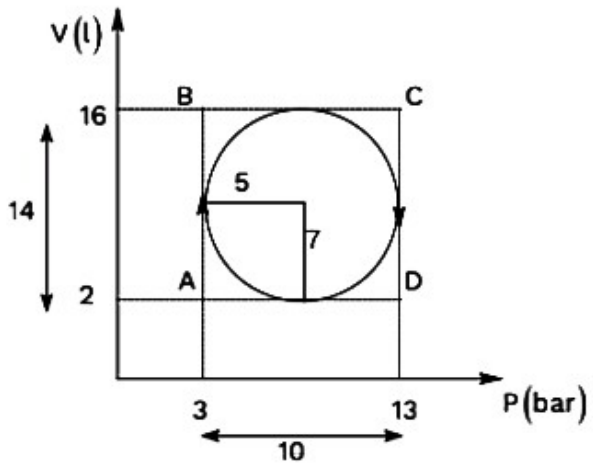
Sol: **A**

$|w|$ = Area of the ellipse

$= \pi a.b$

$= \frac{22}{7} \times 5 \times 7 \text{ bar.L}$

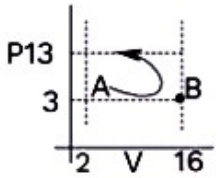
$= 110 \text{ bar L}$



$\because$ anticlockwise $\rightarrow$ w = +110bar⊥

$$= \frac{110 \times 100}{1000} kJ$$

$$= 11kJ$$



w = 11kJ

Work done on the gas = 11 kJ

Work done by the gas = - 11 kJ

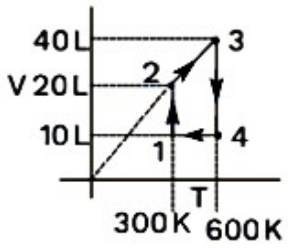
Suppose :

w = - 20J

Work done by the gas = 20 kJ

Work done on the gas = - 20 kJ

Q12 **What is network done (in calories) by 1 mole of monoatomic ideal gas in a process described by 1,2,3,4 in given V-T graph.**



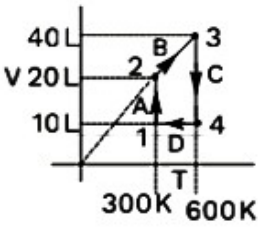
Use: R = 2cal/mole K
ln 2 = 0.7
(A) -600 cal **(B) -660 cal** **(C) + 660 cal** **(D) + 600 cal**

Sol: **B**

$|W_{total}| \neq$ Area of trapesium

(Because not PV graph)

$$w_T = w_A + w_B + w_C + w_D$$



$$w_A = -nR\,300\,\ln\frac{20}{10}$$

$$w_B = -nR\,\Delta T = -nR(600 - 300)$$

$$w_C = -nR\,600\ln\frac{10}{40}$$

$$w_D = 0$$

$$W_T = -nR\left[300\ln 2 + 300 + 600\ln\frac{1}{4}\right]$$

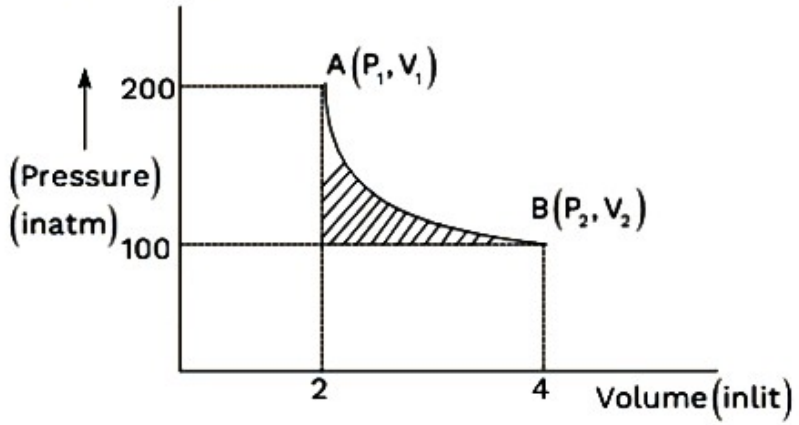
$$= -1\times 2\times 300\left[0.7 + 1 - 2(1.4)\right]$$

$$= -600(1.7 - 2.8)$$

$$= +660\,\text{cal}$$

i.e., work done by gas = -660 cal.

Q13 **An ideal gas change state from A to state B. find work done by gas (in kJ) using the following P-V diagram**



Sol: $\because P_1V_1 = P_2V_2 \Rightarrow$ Isothermal

$\because$ Graph $\Rightarrow$ Reversible isothermal

So,

$$nRT = P_1V_1$$

$$= P_2V_2$$

$$w = -nRT \ln\frac{V_2}{V_1}$$

$$= -400 \ln\frac{4}{2} \text{ atm.L}$$

$$= -280 \text{ atm.L}$$

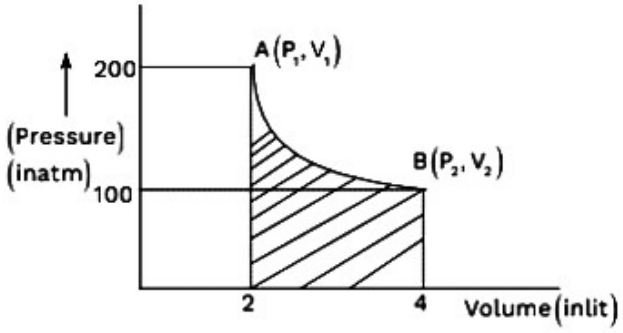
$$w = -280 \times 101.3 \text{ J}$$

$$= -\frac{280 \times 101.3}{1000} \text{ kJ}$$

$$= -28.3 \text{ kJ}$$

Work done by gas = 28.3 kJ

Nearest integer = 28



Heat (q):

A random form of energy exchange between system & surrounding.

Calculation of 'q'

(i) In heating/cooling: (Temperature is changing)

$$C = \frac{dq}{dT}$$

$$C_s = \frac{dq}{M.dT}$$

$$C_m = \frac{dq}{ndT}$$

For a substance.

$$\boxed{\begin{aligned} dq &= C.dT \\ &= M.C_s.dT \\ &= n.C_m.dT \end{aligned}}$$

Put given conditions & then 'integrate'.

$$\int_i^f dq = q$$

If isobaric heating/cooling

$$dq \rightarrow dq_p \left(\text{heat at constant pressure}\right)$$

$$dq_p = C_p.dT$$

$$= m.C_{ps}.dT$$

$$= n.C_{p,m}.dT$$

If Isochoric heating/colling $\left(dq \rightarrow dq_v\right)$

$$dq_v = C_v.dT$$

$$= m.C_{vs}dT$$

$$= n.C_{vm}.m.dT$$

(ii) In isothermal process

i.e. $dT = O$

but $\left.\begin{matrix} dq = O \\ dq \neq O \end{matrix}\right\}$ both are possible

$$dq = \left(du - dw\right)$$

Q14 **80 cal of heat is needed to raise the temperature of 4 moles of an ideal gas by 5 K at constant volume. How much heat is needed to raise the temperature of 4 moles of the same ideal gas by 5 K at constant pressure?**
(A) 80 cal **(B) 120 cal** **(C) 40 cal** **(D) 100 cal**

Sol: **B**

$q_v = 80 = n.c_{vm}.\Delta T$

$80 = 4 \times C_{v,m} \times 5$

$C_{v,m} = 4 \text{ cal / mol.K}$

$C_{p,m} = C_{v,m} + R = 6 \text{ cal / mol K}$

$q_p = n.C_{p,m}.\Delta T$

$= 4 \times 6 \times 5 = 120 \text{ cal}.$

Calculation of ΔU:

First law: dU = dq + dw

constant volume

$dw = 0$

Also $dq \rightarrow dq_v$

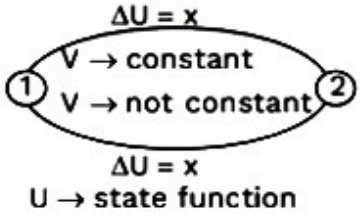
$dU = dq_v + O$

$\Rightarrow dU = Cv.dT$

$$\boxed{C_v = \left(\frac{dU}{dT}\right)_v}$$

also $\boxed{\Delta U = q_v}$

* Means ΔU is equal to the heat exchanged when the process is carried out at constant volume.



U → state function

i.e. If 10 J heat is absorbed at constant volume

$\Rightarrow \Delta U = +10\,J$

If 25 cal heat released at constant volume

$\Rightarrow \Delta U = -25cal.$

Also, in general,

$U = F(T.V)$

So, $dU = \left(\frac{\partial U}{\partial T}\right)_V .dT + \left(\frac{\partial U}{\partial V}\right)_T dV$

$$\boxed{\Rightarrow du = C_v.dT + \left(\frac{\partial U}{\partial V}\right)_T dV}$$

↓ Dependency on Temp. ↓ Dependency on Vol.

For ideal gas	For solids, liquids, and real gases
$U = f(T)$ only $\Rightarrow \boxed{dU = C_v.dT}$ always	$U = F(T, V)$ $\Rightarrow dU = C_v.dT + \left(\frac{\partial U}{\partial V}\right)_T .dV$ If the volume is constant $dV = 0$ $\Rightarrow \boxed{dU = C_v.dT}$ If the volume is not constant $dv \neq 0$ $\Rightarrow \boxed{dU \neq C_v.dT}$

Q15 **For a real gas, determine ΔU if attractive forces (A.F.) are dominant.**

Sol:

In expansion	In compression
i.e., vol ↑ $\Rightarrow dV > 0, \partial V > 0$ also IMD ↑ $\Rightarrow$ A.f. ↓ $\Rightarrow (P.E)_{particle}$ ↑ $\Rightarrow U \uparrow$ $\Rightarrow \partial U > 0$ so $\left(\frac{\partial U}{\partial V}\right) dV > 0$ $\Rightarrow \boxed{dU > C_v.dT}$	i.e., vol ↓ $\Rightarrow dV < 0, \partial V < 0$ also IMD ↓ $\Rightarrow$ A.f. ↑ $\Rightarrow (P.E)_{particle}$ ↓ $\Rightarrow U \downarrow$ $\Rightarrow \partial U < 0$ so $\left(\frac{\partial U}{\partial V}\right) dV < 0$ $\Rightarrow \boxed{dU < C_v.dT}$

Q16 **For a real gas, determine ΔU if repulsive force (R.F.) are dominant.**

Sol:

In expansion	In compression
i.e., vol ↑ $\Rightarrow \partial V > 0, dV > 0$ also IMD ↑ $\Rightarrow$ R.F. ↓ $\Rightarrow (P.E)_{particle}$ ↓ $\Rightarrow U \downarrow$ $\Rightarrow \partial U < 0$ so $\left(\frac{\partial U}{\partial V}\right)_T dV < 0$ $\Rightarrow \boxed{dU < C_v.dT}$	

ΔU the calculation for an ideal gas:

$dU = C_v.dT$ always

$= M.C_{v,s}.dT$

$= n.C_{v,m}.dT$

Let for n mole gas

$dU = n.C_{v,m}.dT$

$$\Delta U = \int_{T_1}^{T_2} dU$$

If $c_{v,m}$ temperature independent	If $C_{v,m}$ temperature dependent
i.e. $C_{v,m}$=constant $\int dU = n, C_{v,m} \int dT$ $\Rightarrow \Delta U = n, C_{v,m}, \Delta T$	Only if given in the question i.e. $C_{v,m} = f(T)$ $\Delta U = n\int_{T_1}^{T_2} C_{v,m}.dT$

Q17 **Statement-I: The internal energy of a closed system always remains constant in an isothermal process.**
Statement-2: The internal energy of an ideal gas is a function of temperature only.
(A) Statement-I is true, statement-2 is true, and statement-2 is correct explanation for statement-I.
(B) Statement-I is true, statement-2 is is true, and statement-2 is NOT the correct explanation for statement-I.
(C) Statement-I is true, statement-2 is false.
(D) Statement-I is false, statement-2 is true.

Sol: **(D)**

Statement-I:

For isothermal process dT = 0

For closed system volume = constant is not compulsory

So internal energy always remains constant is not appropriate.

Statement-II:

The internal energy of an ideal gas is a function of temperature only.

Q18 **5 moles of an ideal gas at 100 K are allowed to undergo reversible compression till its temperature become 200 K. If $C_v = 28\,jK^{-1}mol^{-1}$ calculating ΔpV for this process $\left(R = 8.0\,jK^{-1}mol^{-1}\right)$**

(A) $\Delta U = 2.8kJ : \Delta(PV) = 0.8kJ$ **(B)** $\Delta U = 14kJ : \Delta(PV) = 4kJ$

(C) $\Delta U = 14J : \Delta(PV) = 0.8kJ$ **(D)** $\Delta U = 14kJ : \Delta(PV) = 18kJ$

Sol: **B**

$$dU = n.C_{v,m}, dT$$

$$\Rightarrow \Delta U = n, C_{v,m}\left(T_2 - T_1\right)$$

$$= 5 \times 28\left(200 - 100\right)$$

$$= 14\,kJ$$

$$\Delta(PV) = \Delta(nRT)$$

$$= nR\Delta T$$

$$= 5 \times 8 \times 100$$

$$= 4\,kJ$$

Q19 **For moles of a diatomic gas that is heated from 0°C to 50°C, find the heat supplied to the has if work done by it is zero.**

(A) 780 R **(B) 500 R** **(C) 100 R** **(D) 650 R**

Sol: **B**

$$w = 0 \quad \Rightarrow \quad q = \Delta U$$

$$= n.C_{v,m}.\Delta T$$

$$= 4 \times \frac{5R}{2} \times 50$$

$$= 500R$$

Enthalpy (H)

A state function & an extensive property.

For a system at a state.

$$\boxed{H = U + PV}$$

$$\downarrow \quad \downarrow$$

$$f(T,V) \quad f(P,V)$$

In general

$$\boxed{H = F(T,P)}$$

For ideal

$$H = U + (nRT)$$

$$\downarrow \quad \downarrow$$

$$f(T) \quad f(T)$$

i.e. $\boxed{(H)_{ideal} = F_{(T)}\text{only}}$

* Absolute value of if at a particular state can not be calculated. We can only determine the change in H between two states.

Calculation of ΔH :

1st law: $dU = dq + dw$

constant pressure

$dW = -P.dV$

$dq \rightarrow dq_p$

$dU = dq_p - P.dv$

$(dU + P.dV)_p = dq_p$

$H = U + PV$

$dH = dU + d(PV)$

at constant pressure: $(dU + Pdv) = dH$

i.e. $dH = dq_p \quad \Rightarrow \quad dH = C_p.dT$

$$\Rightarrow \boxed{C_p = \left(\frac{dH}{dT}\right)_p}$$

also $\boxed{\Delta H = q_p}$

Means ΔH are equal to heat exchanged of that process which is carried out at constant pressure.

i.e., If 15 J heat absorbed at constant pressure

$\Rightarrow \Delta H = +15\,J$

If 54 cal. the heat released at constant pressure

$\Rightarrow \Delta H = -54\,Cal$

Also, in general

$$H = f(T,P)$$

$$dH = \left(\frac{\partial H}{\partial T}\right)_P .dT + \left(\frac{\partial H}{\partial P}\right)_T .dP$$

$$\Rightarrow \boxed{dH = C_p.dT + \left(\frac{\partial H}{\partial P}\right)_T .dP}$$

↓ ↓

Temperature dependency | Pressure dependency

For Ideal gas	For solids/liquid/Real gases
$H = f(T)$ only so $\boxed{dH = C_p.dT}$ always	$H = f(T.P)$ $\Rightarrow dH = C_p.dT + \left(\frac{\partial H}{\partial P}\right)_T .dP$ If pressure is constant: $dP = 0$ $\Rightarrow dH = C_p.dT$ If pressure is not constant $dP \neq 0$ $\Rightarrow \boxed{dH \neq C_p.dT}$

For ideal gas always

$PV = nRT$

$C_{p,m} - C_{v,m} = R$

$dU = C_v dT$

$dH = C_p.dT$

ΔH calculation for ideal gas:

$\because$ always, $dH = C_p.dT$

$= MC_{p.s}.dT$

$= n.C_{p.m}.dT$

Let for n mole,

$dH = n.C_{p,m}.dT$

$\Rightarrow \Delta H = \int_{T_1}^{T_2} dH$

If $C_{p,m}$ temperature independent	If $C_{p,m}$ temperature dependent
$C_{p,m}$ =constant $\int dH = n.C_{p,m}\int dT$ $\Rightarrow \boxed{\Delta H = n.C_{p,m}.\Delta T}$	$C_{p,m} = F(T)$ So, $\Delta H = n\int_{T_1}^{T_2} C_{p,m}.dT$

Q20 **For silver, $C_p\left(JK^{-1}mol^{-1}\right) = 23 + 0.01T$. If the temp.(T) of 3 moles of silver(Ag) is increased from 300K to 1000K at 1 atm pressure, the value of ΔH will be close be:**

(A) 16 kJ **(B) 13 kJ** **(C) 21 kJ** **(D) 62 kJ**

Sol: **D**

$C_{p,m} = F(T)$

silver $\rightarrow$ solid

$$dH = C_p.dT + \left(\frac{\partial H}{\partial P}\right)_T .dP$$

at constant pressure

$dH = n.C_{p,m}.dT$

$$\Delta H = 3\int_{300}^{1000} (23 + 0.01T)dT$$

$$= 3\times 23\int_{300}^{1000} dT + 3\times 10^{-2}\int_{300}^{1000} T.dT$$

$$= 69\left(T_2 - T_1\right) + \frac{3\times 10^{-2}}{2}\left(T_2^2 - T_1^2\right)$$

$$= 69\left(1000 - 300\right) + \frac{3\times 10^{-2}}{2}\left(10^6 - 9\times 10^4\right)$$

$$= 69\times 700 + \frac{3\times 10^{-2}}{2}\times 91\times 10^4$$

$$= 48300 + 13650 = 61950\ J = 61.9 \cong 62\ kJ$$

Q21 **5 moles of an ideal gas at 300 K, expanded isothermally from an initial pressure of 4 atm to a final pressure of 1 atm. Against a constant ext. Pressure of 1 atm. Calculate q, w, ΔU & ΔH**

Sol:

$dU = C_v.dT$

$\Delta U = 0$

$dH = C_p.dT$

$\Rightarrow \Delta H = 0$

$\Delta U = q + w$

$q = -w$

$$w_{irr} = \left(-P_{ext}\left(\frac{1}{P_2} - \frac{1}{P_1}\right)\right)nRT$$

$$= -1\left(\frac{1}{1} - \frac{1}{4}\right)\times 5\times R\times 300$$

$$= -1125R$$

Q22 **Calculate values of all if the process in the previous question is carried out reversibly –**

Sol:

$dU = C_v.dT$

$\Delta U = 0$

$dH = C_p.dT$

$\Rightarrow \Delta H = 0$

$\Delta U = q + w$

$q = -w$

$$w_{rev} = -nRT\ln\frac{P_1}{P_2}$$

$$= -5R.300\ln\frac{4}{1}$$

$$= -3000\ln 2R$$

Relation between ΔH and ΔU:

In general,

$\because H = U + PV$

$\Rightarrow \boxed{dH = dU + d(PV)}$

For all systems in all process

If pressure is constant

$dH = dU + P.dV$

$\Rightarrow \boxed{\Delta H = \Delta U + P.\Delta V}$

If the volume is constant

$dH = dU + V.dP$

$\Rightarrow \boxed{\Delta H = \Delta U + V.\Delta P}$

If pressure & volume are not constant

$$\Rightarrow \boxed{\begin{aligned}\Delta H &= \Delta U + \Delta(PU) \\ &= \Delta U + (P_2V_2 - P_1V_1)\end{aligned}}$$

For ideal gas:

$\because PV = nRT$

$dH = dU + d(nRT)$

(i) For heating/cooling/expansion/compression

$dH = dU + nR.dT$

$\Rightarrow \boxed{\Delta H = \Delta U + nR\Delta T}$

$nC_{p,m}dT = n.C_{v,m}.dT + nRdT$

(ii) For reactions/phase transitions involving ideal gas:

It always takes place at a constant temperature

So, $dH = dU + R.T\,dn_g$

$\Rightarrow \boxed{\Delta H = \Delta U + \Delta n_g.RT}$

Points to remember

$\Rightarrow$ Put $\Delta H, \Delta U$ & R in the same unit system

$\Rightarrow$ Temperature must be kelvin

$\Rightarrow$ Since $\Delta H, \Delta U$ & Δng is extensive properties so. Put for the same number of moles.

$N_2O_4(g) \rightarrow 2NO_2(g)$; $\Delta H = x$

$\Delta n_g = (2-1) = 1$ $\Delta U = y$

but $2N_2O_4(g) \rightarrow 4NO_2(g)$

$\Delta n_g = 4 - 2 = 2$

also $\Delta H = 2x$

$\Delta U = 2y$

Q23 **Consider the reaction: $N_2 + 3H_2 \rightarrow 2NH_3$ carried out at constant temperature. If ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true?**

(A) $\Delta H = \Delta U$ **(B)** $\Delta H = 0$ **(C)** $\Delta H > \Delta U$ **(D)** $\Delta H < \Delta U$

Sol: **D**

$\Delta H = \Delta U + \Delta n_g RT$ ($\because \Delta n_g \rightarrow$ negative)

$\Rightarrow \Delta H < \Delta U$

Q24 **$(\Delta H - \Delta U)$ for the formation of carbon monoxide (CO) from its elements at 298 K is – (R=8.314 j K^{-1} mol^{-1})**

(A) $1238.78\,J mol^{-1}$ **(B)** $-2477.57\,J mol^{-1}$

(C) $2477.57\,J mol^{-1}$ **(D)** $-1238.78\,J mol^{-1}$

Sol: **A**

$$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow 1CO_{(g)}$$

$$(\Delta H - \Delta U) = \Delta n_g . RT$$

$$= \left(1 - \frac{1}{2}\right) \times 8.314 \times 298$$

$$= \frac{1}{2} \times 8.314 \times 298$$

$$= 1238.78 \text{ J mol}^{-1}$$

Q25 **For the reaction, $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$: $\Delta H = -560$ kJ mol^{-1}. In one liter vessel at 500 K, the initial pressure is 70 atm, and after the reaction, it becomes 40 atm at a constant volume of one liter. Calculate the change in internal energy. All the above gases show significant deviation from ideal behavior. (1 L. atom = 0.1 kJ)**

Sol: $\Delta H \neq \Delta U + \Delta ng.RT$

$\because$ gases are not ideal

$\because dH = dU + d(PV)$

$\Rightarrow \Delta H = \Delta U + V(P_2 - P_1)$

$(-560) = \Delta U + 1(40 - 70) \times 0.1$

$\Delta U = -560 + 3$

$= -557 kJ / mol$

Q26 **Assuming that water vapor is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vaporized at 1 bar pressure and 100°C, (Given: molar enthalpy of vaporization of water at 1 bar and 373 K = 41 kJ mol^{-1} and R = 8.3 J mol^{-1}K^{-1})-**

(A) 4.100 kJ mol^{-1} **(B) 3.7904 kJ mol^{-1}**

(C) 37.904 kJ mol^{-1} **(D) 41.00 kJ mol^{-1}**

Sol: **C**

$H_2O(l) \rightarrow H_2O(g)$

$\Rightarrow \Delta H = \Delta U + \Delta n_g RT$

$$\Delta U = 41 - \frac{(1) \times 8.3 \times 373}{1000} = 37.904 \text{ kJ mol}^{-1}$$

Adiabatic expansion/compression of ideal gas

$dq = o$

so, $dU = dw$

ideal $C_v.dT = dw$

Let $C_v \rightarrow$ temperature independent

$$W_{adiabatic,ideal} = \begin{cases} nC_{V,M}.\Delta T \\ nC_{V,M}(T_2 - T_1) \\ \dfrac{nR(T_2 - T_1)}{\gamma - 1} \\ \dfrac{P_2V_2 - P_1V_1}{\gamma - 1} \end{cases}$$

Points to remember

If "reversible adiabatic process" and an "irreversible adiabatic process" are started from the same initial state, their final states can never be the same.

It is applicable for both reversible and irreversible processes.

Case I: Reversible adiabatic

$dU = dq_{rev.} + dw_{rev.}$

$dU = -P_{gas}.dV$

Ideal gas

$$n.C_{v.m}.dT = -\frac{nRT}{V}.dV$$

$$n\frac{R}{\gamma-1}dT = -\frac{nRT}{V}dV$$

$$\int_{T_1}^{T_2}\frac{1}{T}.dT = -(\gamma-1)\int_{V_1}^{V_2}\frac{1}{V}.dV$$

$$\ln\frac{T_2}{T_1} = -(\gamma-1)\ln\frac{V_2}{V_1}$$

$$\ln\frac{T_2}{T_1} = \ln\left(\frac{V_2}{V_1}\right)^{-(\gamma-1)}$$

$$\left(\frac{T_2}{T_1}\right) = \left(\frac{V_2}{V_1}\right)^{-(\gamma-1)}$$

$$\left(\frac{T_2}{T_1}\right)\left(\frac{V_2}{V_1}\right)^{\gamma-1} = 1$$

i.e., $T.V^{\gamma-1}$ = constant

for ideal $T \propto (P.V.)$

$(P.V).V^{\gamma-1}$ = constant

$\Rightarrow$ $P.V^{\gamma}$ = constant

For ideal $V \propto \left(\frac{T}{P}\right)$

So, $P.\left(\frac{T}{P}\right)^{\gamma}$ = constant

$\Rightarrow$ $T^{\gamma}.P^{1-\gamma}$ = constant

Only for the ideal gas in reversible adiabatic process.

Case II: Irreversible adiabatic-

$dU = dq_{irr.} + dw_{irr.}$

Ideal $nC_{C,M}.dT = -P_{ext}.dV$

Let, $C_{V,M}$ is temperature independent

$$nC_{V,M}.(T_2-T_1) = -P_{ext}.(V_2-V_1)$$

$$= -P_{ext}.\left(\frac{nRT_2}{P_2}-\frac{nRT_1}{P_1}\right)$$

PV^{γ} = constant, is not applicable of irreversible adiabatic process.

Points to remember

for adiabatic, ideal gas

If reversible $\Rightarrow$ PV^{γ} = Constant

If irreversible $\Rightarrow$ 1st law

Q27 **An ideal diatomic gas was allowed to expand reversibly and adiabatically from 1 liter to 32 liters. Its initial temperature was 327°C. The molar enthalpy change (in J/mole) for the process is :**

(A) –1125 R **(B) –575 R**

(C) –1575 R **(D) None of these**

Sol: **(C)**

Reversible and adiabatic

$\Rightarrow PV^{\gamma} = K$

$TV^{\gamma-1} = \text{constant}$

$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$= 600\left(\frac{1}{32}\right)^{2/5}$$

$$= 150\ K$$

Ideal :

$dH = nC_{p,m}.dT$

$\Rightarrow \Delta H = n.C_{p.m.}(T_2 - T_1)$

$$= 1 \times \frac{7R}{2}(150 - 600)$$

$$= -1575R$$

Q28 **At initial temperature T, One mole of a monoatomic ideal gas expands adiabatically against a constant external pressure of 1 atm from one liter to two liters. Find out the final temperature (R = 0.0821 liters. atm K^{-1} mol^{-1})**

(A) T **(B)** $\dfrac{T}{(2)^{\frac{5}{3}-1}}$ **(C)** $T - \dfrac{2}{3\times 0.0821}$ **(D)** $T + \dfrac{2}{3\times 0.0821}$

Sol: **(C)**

Against constant $P_{ext.} \Rightarrow$ irreversible $\Rightarrow 1^{st}$ law,

PV^{γ} = constant not applicable

$dU = 0 + dW_{irr.}$

$n.C_{v,M}(T_2 - T_1) = -P_{ext}(V_2 - V_1)$

$$1 \times \frac{3R}{2}(T_2 - T_1) = -1(2-1)$$

$$T_2 = T_1 - \frac{2}{3 \times 0.0821}$$

Q29 **One mole of a liquid (1 bar, 100ml) is taken in an adiabatic container, and the pressure rises sharply to 100 bar, and at a constant pressure of 100 bar, the volume reduces by 1 ml. Find ΔU and ΔH.**

Sol: $\Rightarrow$ irreversible adiabatic for liquid

(1bar, 100 ml) $\rightarrow$ (100 bar, 100 ml) $\rightarrow$ (100 bar, 99 ml)

$q = 0$

volume constant $\Rightarrow W = 0$

$\Rightarrow \Delta U_1 = 0$

$\therefore$ $dH = dU + d(PV)$

$\Delta H_1 = 0 + V(P_2 - P_1)$

$= 100(100-1)$ bar ml

$= 9900$ bar.ml

$q = 0$

$W_{irr} = -P_{ext}.\Delta V$

$= -100\ (99-100)$

$= +100$ bar.ml

$\Delta U_2 = 0 + W$

$= +100$ bar.ml

$dH = dU + d(PV)$

$\Delta H_2 = \Delta U_2 + P\Delta U$

$= 100 + 100(99-100)$

$= 0$

So, Total

$\Delta U = 0 + 100$ bar.Ml

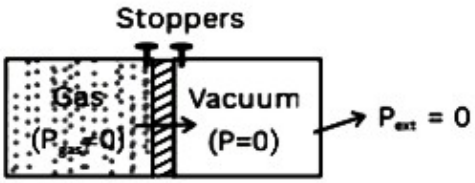
$= 100 \times 10^{-3}$ bar.L

$= 100 \times 10^{-3} \times 100$ J

$\Delta H = 9900 + 0 = 9900 \times 10^{-3} \times 100$J

$= 990$ J

Free expansion: Expansion against vacuum



When stoppers are removed, the piston will always move rightwards always means expansion.

A free expansion is always an irreversible process.

$$dW = -P_{ext.}\,dV$$
$$\downarrow$$
$$0$$

$\Rightarrow$ dW = 0 for both ideal and non-ideal gas.

For ideal gas

Isothermal free expansion	Adiabatic free expansion
$dT = 0 \Rightarrow dU = C_v\,dT$ $\Rightarrow \Delta U = 0$ also, $dH = C_p\,dT$ $\Rightarrow \Delta H = 0$ 1st Law, $dU = dq + dW$ $\Rightarrow dq = 0$ adiabatic	$dq = 0$ also $dW = 0$ 1st law, $dU = 0$ ($dU = C_V dT$) $\Rightarrow dT = 0$ (Isothermal) Also, $\Delta U = 0$ and $\Delta H = 0$

Polytropic process :

If gives a generalized equation for an ideal gas which is subjected to reversible change:

PV^x = constant

x = real number

for each values of x there will be a unique reversible process.

If x = 0 $\Rightarrow$ reversible isobaric

x = 1 $\Rightarrow$ reversible isothermal

x= γ $\Rightarrow$ Reversible adiabatic

x = ∞ $\Rightarrow$ Reversible isochoric

Molar Heat capacity in polytropic process (C_M)

$$dU = dq_{rev.} + dw_{rev.}$$

Points to remember

This Means free expansion of ideal gas is isothermal and adiabatic simultaneously.

Ideal :

$n.C_{v,M}.dT = n.C_M.dT - P_{gas}dV$

$C_M = C_{v,M} + \frac{P}{n}\left(\frac{dV}{dT}\right)$

Put $\left(\frac{dV}{dT}\right)$ from the polytropic process

For $PV^x = K$

$\because$ $PV = nRT$

$\frac{K}{V^x}.V = nRT$

$K.V^{1-x} = nRT$

$K.(1-x).V^{1-x-1}\,dV = nRdT$

$\frac{K}{V^x}(1-x).dV = nRdT$

$P(1-x)dV = nR.dT$

$\Rightarrow \frac{P}{n}\left(\frac{dV}{dT}\right) = \frac{R}{1-x}$

Put,

$\Rightarrow C_M = C_{v,M} + \frac{R}{1-x}$

For ideal gas in $PV^x = K$

The molar heat capacity of a monoatomic gas for which the ratio of pressure and volume is one.

Q30 **(A) 4/2 R** **(B) 3/2 R** **(C) 5/2 R** **(D) Zero**

(A)

monoatomic gas = $C_{v,M} = \frac{3R}{2}$

Sol: $\frac{P}{V} = 1(\text{constant})$

$\Rightarrow PV^{-1} = \text{constant}$

i.e. $x = (-1)$

So, $C_M = \frac{3R}{2} + \frac{R}{1-(-1)} = 2R$

Some important comparisons for an ideal gas:

1. $|w_{rev}|$vs$|w_{irr}|$ for same V_1 to V_2

In isobaric process for same V_1 to V_2 :

$|W_{rev}| = |W_{irr}|$

In other processes for same V_1 to V_2 :

If expansion

$|W_{rev}| > |W_{irr}|$

If compression

$|W_{irr}| > |W_{rev}|$

Case- I :

Expansion (let isothermal)

Single-step expansion :

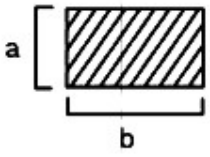
i.e. $P_1V_1 \xrightarrow{\text{irr.}} P_2V_2$

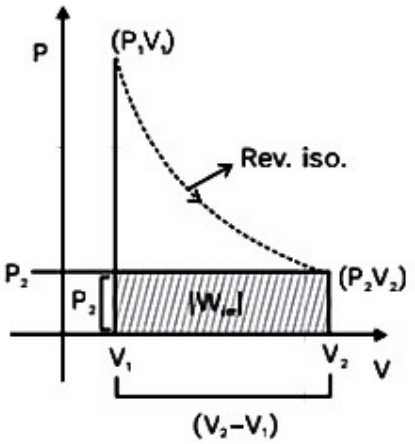
$W_{irr} = -P_{ext}\Delta V$

$= -P_2(V_2-V_1)$

$|W_{irr}| = P_2 \times (V_2-V_1)$

$a \times b$





Two step expansion :

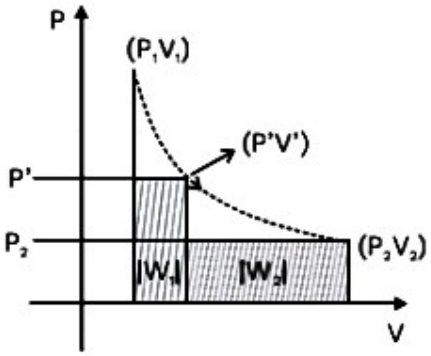
$P_1V_1 \xrightarrow{I} P'V' \xrightarrow{II} P_2V_2$

$W_1 = -P'(V'-V_1)$

$\Rightarrow |W_1| = P' \times |V'-V_1|$

Also, $W_2 = -P_2(V_2-V')$

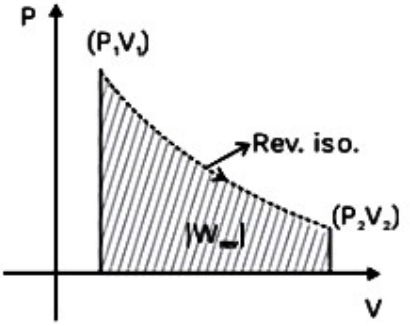
$\Rightarrow |W_2| = P_2 \times |V_2-V'|$



∞-step expansion:

i.e. Reversible Process

$|W_{rev}|$ = Total area under the curve



Case II :

Compression (let isothermal)

Single step compression:

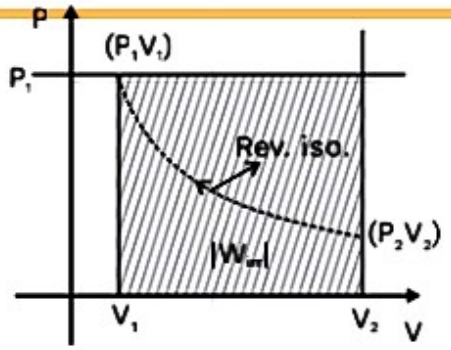
$P_2V_2 \longrightarrow P_1V_1$

$\Rightarrow$ Irreversible

i.e. $W_{irr} = -P_{ext.}\Delta V$

$= -P_1(V_1-V_2)$

$|W_{irr}| = P_1 \times |V_1-V_2|$



Two-step compression:

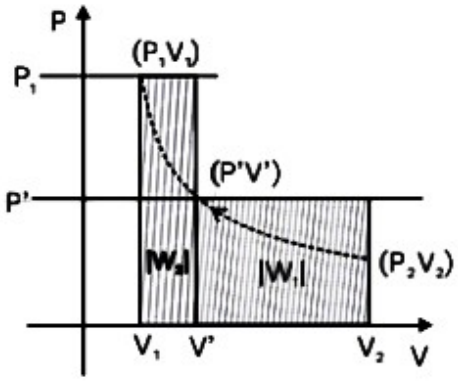
$P_2V_2 \xrightarrow{W_1} P'V' \xrightarrow{W_2} P_1V_1$

$W_1 = -P'(V'-V_2)$

$|W_1| = P' \times |V'-V_2|$

$W_2 = -P_1(V_1-V')$

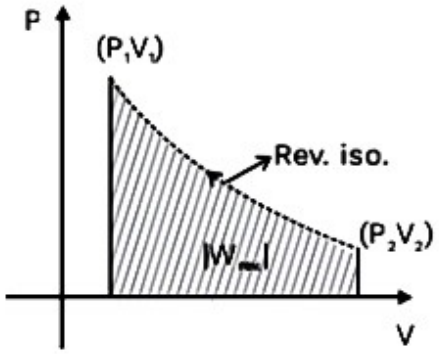
$\Rightarrow |W_2| = P_1 \times |V_1-V'|$



∞-step compression

$\Rightarrow$ Reversible process

$|W_{rev.}|$ = total area under the curve



2. Reversible adiabatic V/S Reversible isothermal-

For ideal gas-

If reversible isothermal $\Rightarrow PV$ = constant

If reversible adiabatic $\Rightarrow PV^{\gamma}$ = constant ($\gamma > 1$)

So, if P v/s V graphs are plotted, then both curves are hyperbolic with different slopes.

$$\text{Slope} = \left(\frac{dP}{dV}\right)$$

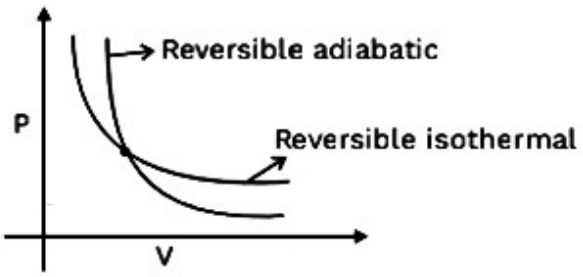
For reversible isothermal

$$\text{Slope} = \left(-\frac{P}{V}\right)$$

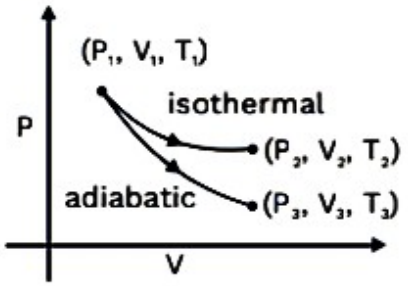
For reversible adiabatic

$$\text{Slope} = \left(-\gamma.\frac{P}{V}\right)$$

$|\text{Slope}|_{\text{adiabatic}} > |\text{slope}|_{\text{isothermal}}$



Q31 **Which of the following statement(s) is (are) correct?**
The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure.



(A) $T_1 = T_2$
(B) $T_3 > T_1$
(C) $W_{isothermal} > W_{adiabatic}$
(D) $\Delta U_{isothermal} > \Delta U_{adiabatic}$

Sol: **(AD)**
(A) $T_1 = T_2$
(B) $dq = 0$
(C) $|W_{iso}| > |W_{adia}|$
(D) $\Delta U_{isothermal} > \Delta U_{adiabatic}$
Zero > (–)ve
So, $C_v dT = dW$
$\Rightarrow$ $dT < 0$
$\Rightarrow$ $T_3 < T_1$

3. Final temperature in adiabatic process (for ideal gas only)

$dq = 0$
$\Rightarrow$ $dU = 0 + dW$
For Ideal gas-
$C_v dT = dW$
If compression $\Rightarrow dW > 0 \Rightarrow dT > 0 \Rightarrow$ temperature$\uparrow$
If expansion $\Rightarrow dW < 0 \Rightarrow dT < 0 \Rightarrow$ temperature$\downarrow$
i.e. $(T_f)_{irr} > (T_f)_{rev.}$
$\because$ $C_v dT = dW$
$\Rightarrow$ $|\Delta T| \propto |W|$

In expansion :
$\because$ $|W_{rev}| > |W_{irr}|$
$\Rightarrow$ $|\Delta T_{rev}| > |\Delta T_{irr}|$

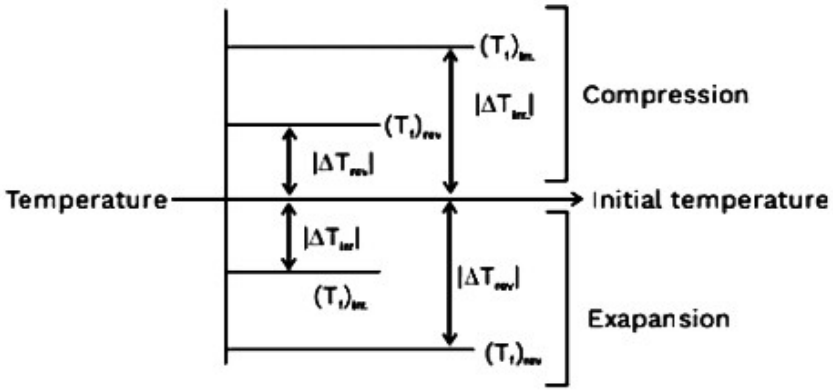
In compression:
$\because$ $|W_{irr}| > |W_{rev}|$

Points to remember

Means in adiabatic expansion/ compression of an ideal gas, the temperature is **always** changed. (except free expansion)

Whether adiabatic expansion or compression of an ideal gas, the final temperature is always greater for the irreversible path.

$\Rightarrow |\Delta T_{irr}| > |\Delta T_{rev}|$



Limitations of 1st law:

1. Is it possible for a system to convert heat into work completely.
2. Whether a process will be spontaneous or not is give conditions.
3. Upto what extent a chemical reaction will proceed in given conditions.

Entropy(s)

Entropy (S) is a state function, and it is an extensive property.

For a system, entropy can be defined as the measure of randomness/ disorderness/disturbance among particles, as follows:

More randomness
or
more disorderness } more entropy
or
more disturbance

And vice-versa.

Comparisons of entropy :

1. On the basis of physical state :
   $(S)_{Solid} < (S)_{liquid} << (S)_{gas}$
   also for a particular substance :
   $(S)_{pure} < (S)_{mix}$

Pure	Mix
1 mole $O_2(g)$	1 mole O_2 + NH_3

2. For solids :
   $(S)_{crystalline} < (S)_{amorphous}$
3. For substances made of similar elements:
   Under similar conditions
   more particle size $\Rightarrow$ more atomicity $\Rightarrow$ more molar mass $\Rightarrow$ more entropy

Example : $O_{(g)} < O_{2(g)} < O_{3(g)}$

$CO_{2(g)} > CO_{(g)}$

$H_2O_{(l)} < H_2O_{2(l)}$

$H_2O_{(g)} > H_2O_{2(l)}$

4. For chemical reactions and phase transitions:

(i) If $\Delta n_g > 0 \Rightarrow S\uparrow \Rightarrow \Delta S > 0$

Example : $N_2O_{4(g)} \longrightarrow 2NO_{2(g)}$

1 mole gas $\longrightarrow$ 2 mole gas

(ii) If $\Delta n_g < 0 \Rightarrow S\downarrow \Rightarrow \Delta S < 0$

Example : $N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$

4 mole gas $\longrightarrow$ 2 mole gas

(iii) If $\Delta n_g = 0 \Rightarrow \Delta S = 0$ & $\Delta S \neq 0$ both are possible.

Example : $CO_{(g)} + H_2O_{(g)} \longrightarrow CO_{2(g)} + H_{2(g)}$

2 mole gas $\longrightarrow$ 2 mole gas

$\Delta n_g = 0$

But for ΔS, data should be given

Example : $H_2O_{(s)} \longrightarrow H_2O_{(l)}$

less (s) More (s)

$\Rightarrow$ $S\uparrow$

$\Rightarrow$ $\Delta S > 0$

Spontaneous process:

The process which proceeds without any external processor, which places on its own under given conditions, is known as a spontaneous process.

It is also known as self occurring process.

E.g. Mixing of gases.

Melting of ice at room Temperature and Pressure.

Heat flow from high-temperature → low-temperature.

Non-spontaneous process:

The process which can not take place on its own is known as a non-spontaneous process.

This means a non-spontaneous process can only take place by the application of some external work on the system.

Points to remember

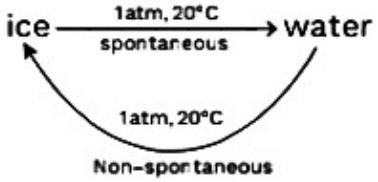
All spontaneous processes are irreversible in nature.

Example:
Compression of gas.
Heat flow from low temperature to high temperature.
(Refrigerator, Ac)

The second law of thermodynamics:

1. Complete conversion of heat into work is not possible.
2. Heat can not flow from low temperature to high temperature without the application of external work.
3. Spontaneous process can not proceed in reverse direction under similar conditions.

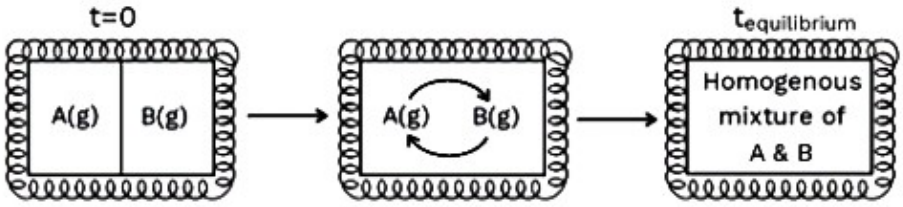
Example:



4. During a spontaneous process, the entropy of the universe increases, and at equilibrium, it remains constant.

Example:

Mixing of gases in isolate system
↓ ↓
Spontaneous Universe



$0 < t < \text{equilibrium}$

$(S)_{sys} = (S)_A + (S)_B$
↑ ↑

$\Rightarrow$ $(S)_{\text{isolated system}}$ increasing in spontaneous process

So, for an irreversible process:

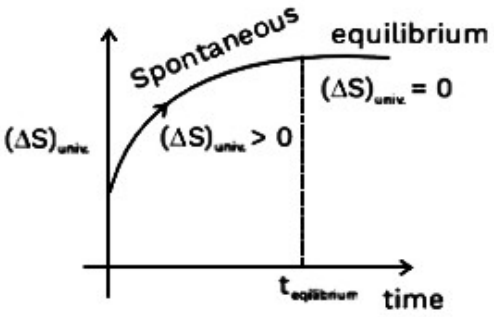
$(\Delta S)_{univ.} \geq 0$ Spontaneity criterion

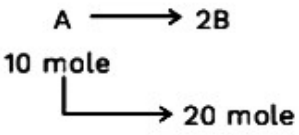
If $(\Delta S)_{univ.} > 0$ Spontaneous

$(\Delta S)_{univ.} = 0$ equilibrium

$(\Delta S)_{univ.} < 0$ non-spontaneous

i.e.,





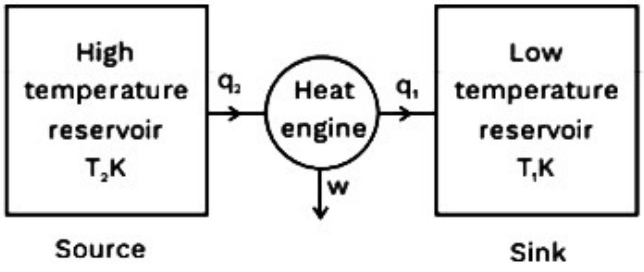
Complete conversion is not possible



Complete conversion possible

Heat engine:

A device that absorbs heat from a source and converts it into work.



$T_2 > T_1$

$\because q_1 \neq 0$

$\Rightarrow w = (q_2 - q_1)$

$< q_2$ (i.e., Partial conversion)

This can also be proved by a heat engine which is based upon the Carnot cycle.

Carnot cycle :

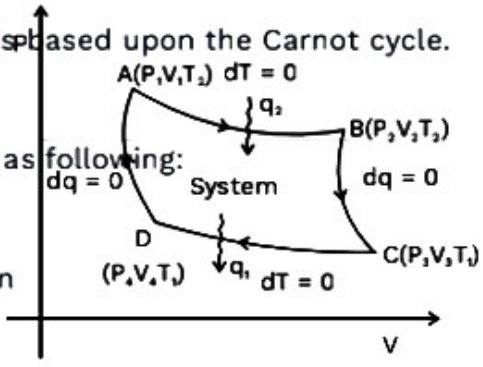
It is a cyclic process consisting of four steps as following:

(i) Reversible isothermal ideal gas expansion

(ii) Reversible adiabatic ideal gas expansion

(iii) Reversible isothermal ideal gas compression

(iv) Reversible adiabatic ideal gas compression



$(\therefore T_2 > T_1)$

Step-1 : $(dT = 0)$

$$W_1 = -nRT_2 \ln \frac{V_2}{V_1}$$

$\therefore$ $\Delta U_1 = q_2 + W_1$ $\quad (\because \Delta U_1 = 0)$

$q_2 = -W_1$

$$= + nRT_2 \ln \frac{V_2}{V_1}$$

Step-2: $(dq = 0)$

$\Rightarrow$ $w_2 = \Delta U_2$

$= n.C_{v,M}(T_1 - T_2)$

Step-3: $(dT = 0)$

$$W_3 = -nRT_1 \ln \frac{V_4}{V_3}$$

Step-4: $(dq = 0)$

$W_4 = \Delta U_4$

$= nC_{v,M}(T_2 - T_1)$

For one complete cycle,

$$(q)_{absorbed} \Rightarrow q_2 = nRT_2 \ln \frac{V_2}{V_1}$$

Also, $W_{total} = W_1 + W_2 + W_3 + W_4$

$$= (-nRT_2 \ln \frac{V_2}{V_1}) + n.C_{v,M}(T_1 - T_2) + (-nRT_1 \ln \frac{V_4}{V_3}) + nC_{v,M}(T_2 - T_1)$$

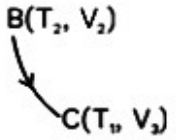
$$= (-nRT_2 \ln \frac{V_2}{V_1}) + n.C_{v,M}(T_1 - T_2) + (-nRT_1 \ln \frac{V_4}{V_3}) + (-nC_{v,M}(T_1 - T_2))$$

$$= (-nRT_2 \ln \frac{V_2}{V_1}) + (-nRT_1 \ln \frac{V_4}{V_3})$$

Reversible adiabatic:

$T_2.V_2^{\gamma-1} = T_1.V_3^{\gamma-1}$

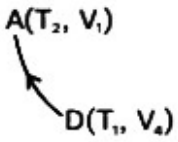
$$\frac{T_2}{T_1} = \left(\frac{V_3}{V_2}\right)^{\gamma-1}$$



Reversible adiabatic :

$T_1.V_4^{\gamma-1} = T_2.V_1^{\gamma-1}$

$$\frac{T_2}{T_1} = \left(\frac{V_4}{V_1}\right)^{\gamma-1}$$



$$\frac{T_2}{T_1} = \frac{T_2}{T_1}$$

$$\left(\frac{V_3}{V_2}\right)^{\gamma-1} = \left(\frac{V_4}{V_1}\right)^{\gamma-1}$$

$\Rightarrow \frac{V_4}{V_3} = \frac{V_1}{V_2}$ put in W_{total}

$$W_{total} = -nRT_2 \ln \frac{V_2}{V_1} - nRT_1 \ln \frac{V_1}{V_2}$$

$$= -nRT_2 \ln \frac{V_2}{V_1} + nRT_1 \ln \frac{V_2}{V_1}$$

So, Work done by the engine = $(-W_{total})$

$$= nR(T_2 - T_1) \ln \frac{V_2}{V_1}$$

$\therefore (q)_{absorbed} = nRT_2 \ln \frac{V_2}{V_1}$

$\Rightarrow (-W_{total}) < q_{absorbed}$

i.e., Partial conversion

The efficiency of a heat engine (η)-

In general:

$$\eta = \left(\frac{\text{work done by engine}}{\text{heat absorbed}}\right) \text{ in the same number of cycles}$$

i.e., $\eta = \frac{|w_{total}|}{q_{absorbed}}$

% efficiency = $\eta \times 100$

So, $\eta = \frac{(q_2 - q_1)}{q_2}$ (put without sign)

$\eta = \frac{(q_2 + q_1)}{q_2}$ (put with sign)

For Carnot engine:

$$\eta = \frac{nR.(T_2 - T_1)\ln \frac{V_2}{V_1}}{nRT_2 \ln \frac{V_2}{V_1}}$$

$$\eta = \frac{T_2 - T_1}{T_2}$$ for carnot engine

$T_2 > T_1$ (in kelvin)

Carnot theorems:

Ist theorem:

Between the same source and sink, the efficiencies of all reversible engines are the same.

$\therefore$ Carnot engine $\Rightarrow$ Reversible process

i.e., $(\eta)_{\text{any rev.}} = \frac{T_2 - T_1}{T_2}$

IInd theorem:

Between the same source and sink, the efficiency of any irreversible engine is lesser as compared to a reversible engine.

i.e., $(\eta)_{\text{any irr.}} < (\eta)_{\text{rev.}}$

$\Rightarrow$ $\eta_{\text{rev.}}$ = maximum

$\eta_{\text{irr}} \neq \left(\frac{T_2 - T_1}{T_2}\right)$

Q32 **A Carnot engine has an efficiency of 40%. Its low-temperature reservoir is at 70°C. What is the temperature (in°C) of the source?**

Sol:

$$\eta = 0.4 = \frac{T_2 - 343}{T_2}$$

$$2T_2 = 5T_2 - 5(343)$$

$$T_2 = \frac{5}{3} \times 343K$$

$$= 571.6K$$

Q33 **The maximum efficiency of a heat engine operating between 100° C and 25°C is:**

(A) 20.11% **(B) 22.2%** **(C) 25.17%** **(D) None**

Sol: **(A)**

$$\eta_{maximum} \Rightarrow h_{reversible} = \frac{T_2 - T_1}{T_2} \times 100$$

$$= \frac{373 - 298}{373} \times 100$$

$$= \frac{75}{373} \times 100$$

$$= 20.11\%$$

Calculation of ΔS:

for infinitesimally small change:

$$ds = \frac{dq_{rev.}}{T}$$

Where,

$dq_{rev.}$ → heat exchanged reversibly (J or cal.)

T →Kelvin temperature (always positive)

dS → Change in entropy $\left(\frac{J}{K} \text{or} \frac{Cal.}{K}\right)$

i.e., If heat absorbed ⇒ dq > 0 ⇒ dS > 0 ⇒ entropy↑

If heat released ⇒ dq < 0 ⇒ dS < 0 ⇒ entropy ↓

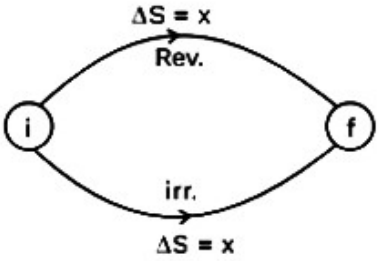
$$\text{Now, } \Delta S = \int_i^f dS$$

$$\text{So, } (\Delta S)_{system} = \int_i^f \left(\frac{dq_{rev.}}{T}\right)_{sys.}$$

$$(\Delta S)_{Surroundings} = \int_i^f \left(\frac{dq_{rev.}}{T}\right)_{surr.}$$

Calculation of $(\Delta S)_{system}$

Since S is a state function, so $(\Delta S)_{system}$ will be the same for both reversible and irreversible paths.



For heating/cooling for substances:

i.e., When temperature of substance is changing

$$\left.\begin{aligned} dq &= C.dT \\ &= m.Cs.dT \\ &= n.C_M.dT \end{aligned}\right\}$$ If isobaric $\Rightarrow C_p$, If isochoric $\Rightarrow C_v$

So, $dS = \int_{T_1}^{T_2} \frac{dq}{T}$

For phase transition:

Always takes place at constant temperature and Pressure.

i.e., $dq \rightarrow dq_p = dH$

So, $dS = \frac{dH}{T}$

$$\int dS = \frac{1}{T}\int dH$$

$$\Rightarrow (\Delta S)_{\text{phase transition}} = \frac{(\Delta H)_{\text{Phase transition}}}{\text{Transition temperature}}$$

Note:

Above relation is applicable only and only if the process is taking place at the transition temperature. (Melting point, boiling point, etc.)

E.g. $H_2O(l) \longrightarrow H_2O(g)$

If T = 100°C (boiling point)

$$\Rightarrow \quad \Delta S_{vap.} = \frac{\Delta H_{vap.}}{373}$$

But if T = 150°C

$$\Delta S_{vap.} \neq \frac{\Delta H_{vap.}}{423}$$

Q34 **Calculate the boiling point of the liquid at 1 atm if the enthalpy of vaporization of a liquid is 30 kJ mol^{-1}, and entropy of vaporization is 75 J/mol.K.:**
(A) 250 K **(B) 400 K** **(C) 450 K** **(D) 600 K**

Sol: **(B)**

$$(\Delta S)_{vap.} = \frac{(\Delta H)_{vap.}}{\text{boiling point}}$$

$$\text{Boiling point} = \frac{30 \times 10^3}{75}$$

$$= 400 \text{ K}$$

For chemical reactions:

$(\Delta S)_{sys} \Rightarrow \Delta_r s$ (entropy change of reaction) $= \sum (S)_{products} - \sum (S)_{reactents}$

$\Rightarrow \Delta_r S = \Sigma(n)_{products}.(S_m)_{product} - \Sigma(n)_{reactant}(S_m)_{reactant}$ $\quad (\because S_m = \frac{S}{n})$

Where, n → moles (stoichiometric coefficient)
S → entropy (J/K)
S_m → molar entropy (J/mol.K)

Example:

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

$\Delta_r S = 2S_m(NH_3, (g)) - 1S_m(N_2, (g)) - 3S_m(H_2, (g))$

Eg. $C_6H_6(l) + \frac{15}{2}O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(l)$

$\Delta_r S = 6S_m(CO_2, (g)) + 3S_m(H_2O, (l)) - 1S_m(C_6H_6, (l)) - \frac{15}{2}S_m(O_2, (g))$

Eg. $H_2O(l) \longrightarrow H_2O(g)$

$\Delta_r S = 1S_m(H_2O, (g)) - 1S_m(H_2O, (l))$

For ideal gas:

1st law: $dU = dq_{rev.} + dw_{rev.}$

$dq_{rev.} = dU - dW_{rev.}$

$= dU - (-P_{gas}.dV)$

$= dU + P.dV$

Now, $\because dS = \frac{dq_{rev}}{T}$

$$= \frac{dU + PdV}{T}$$

$$dS = \frac{n.C_{v,M}.dT}{T} + \frac{nR}{V}dv$$

$$\int_i^f dS = n.C_{v,M}.\int_{T_1}^{T_2}\frac{1}{T}.dT + nR\int_{V_1}^{V_2}\frac{1}{V}.dV$$

$$\Rightarrow \quad (\Delta S)_{ideal} = n.C_{V,M}.\ln\frac{T_2}{T_1} + nR\ln\frac{V_2}{V_1}$$

$$= n.C_{P,M}\ln\frac{T_2}{T_1} + nR\ln\frac{P_1}{P_2}$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{V_2}{V_1} = \left(\frac{T_2}{T_1}\right)\left(\frac{P_1}{P_2}\right)$$

Q35 **In the isothermal reversible expansion, an ideal gas having of 2 moles and volume increases of 10 dm^3 to 100 dm^3 at 27°C, calculate the change in entropy involved:**

(A) 38.3 JK^{-1} **(B) 35.8 JK^{-1}** **(C) 32.3 JK^{-1}** **(D) 42.3 JK^{-1}**

Sol: **(A)**

$T_1 = T_2$

$$\Delta S = nR\ln\frac{V_2}{V_1}$$

$$= 2 \times 8.314 \times \ln\frac{100}{10}$$

$$= 2 \times 8.314 \times 2.303$$

$$= 38.3\ J/K$$

Reversible iso.

$dU = 0$

$$\Delta S = \frac{q_{rev.}}{T}$$

Q36 **Two blocks of the same metal having equal mass, at temperature T_1 and T_2, respectively, are taken in contact with each other and allowed to attain equilibrium at constant pressure. The change in entropy(ΔS) for this process is:**

(A) $C_p \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$

(B) $2C_p \ln\left[\frac{(T_1+T_2)^{1/2}}{T_1T_2}\right]$

(C) $2C_p \ln\left[\frac{T_1+T_2}{4T_1T_2}\right]$

(D) $2C_p \ln\left[\frac{T_1+T_2}{2T_1T_2}\right]$

Sol: **(A)**

at constant pressure



T_1 T_2 $T_F = \frac{T_1+T_2}{2}$

$$\Delta S = \Delta S_1 + \Delta S_2$$

$$= \int_{T_1}^{T_f} \frac{C_p.dT}{T} + \int_{T_2}^{T_f} \frac{C_p.dT}{T}$$

$$= C_p.\ln\frac{T_f}{T_1} + C_p.\ln\frac{T_f}{T_2}$$

$$= C_p.\ln\frac{T_f^2}{T_1.T_2}$$

Calculation of $(\Delta S)_{surroundings}$:

$$\therefore \quad (\Delta S)_{Surrounding} = \int \frac{(dq_{rev.})_{surrounding}}{T_{surrounding}} \; ; \; (dq)_{surrounding} = -(dq)_{sys.}$$

Due to very large size, the surrounding always moves reversibly, whether system moves reversibly or irreversibly.

If reversible process:

$$dq \Rightarrow dq_{rev.} \Rightarrow (dq_{rev})_{surroundings} = -(dq_{rev})_{system}$$

If irreversible process:

$$dq \Rightarrow dq_{irr.} \Rightarrow (-dq_{irr.})_{system} = (dq_{rev})_{surroundings}$$

If reversible process:

$-(dq_{rev.})_{sys.} = (dq_{rev.})_{surr.}$

$$\therefore \quad (\Delta S_{surr.}) = \int \frac{(dq_{rev.})_{surr.}}{T}$$

$$= -\int \frac{(dq_{rev.})_{sys.}}{T}$$

$$= -(\Delta S)_{system}$$

If irreversible process:

$-(dq_{irr.})_{sys.} = (dq_{rev.})_{surr.}$

$$\text{So,} \quad (\Delta S_{surr.}) = \int \frac{(dq_{rev.})_{surr.}}{T_{surr.}}$$

$$= -\int \frac{(dq_{irr.})_{sys.}}{T_{surr.}}$$

$$\Rightarrow \quad (\Delta S)_{surr.} = \frac{-q_{irr.}}{T_{surr.}}$$

Where $(T)_{surr.} = T_2$ of system

Calculation of $(\Delta S)_{univ.}$:

$\because$ System + surrounding = universe

$\Rightarrow \quad (\Delta S)_{universe} = (\Delta S)_{System.} + (\Delta S)_{surrounding}$

Where,

$$(\Delta S)_{sys.} = \int \frac{dq_{rev.}}{T}\text{; (both reversible and irreversible)}$$

$$(\Delta S)_{Surr.} \xrightarrow{\text{reversible}} = (-\Delta S_{sys.})$$

$$(\Delta S)_{Surr.} \xrightarrow{\text{irreversible}} \neq -(\Delta S)_{sys.} = \frac{-q_{irr.}}{T_{surr.}}$$

i.e., For a reversible process:

$(\Delta S)_{surr.} = -(\Delta S)_{sys.}$

$\Rightarrow \quad (\Delta S)_{univ.} = 0$

i.e., equilibrium (Though out the process)

1. In isothermal ideal gas expansion/compression (dT = 0)

(i) If reversible process:

$$(\Delta S)_{sys.} = n.R.\ln\frac{V_2}{V_1} \text{ or } n.R.\ln\frac{P_1}{P_2}$$

$\therefore$ Reversible:

$(\Delta S)_{surr.} = -(\Delta S)_{sys.}$

$\Rightarrow$ $(\Delta S)_{univ.} = 0$

(ii) If irreversible process:

$$(\Delta S)_{sys.} = n.R.\ln\frac{V_2}{V_1} \text{ or } n.R.\ln\frac{P_1}{P_2}$$

$\because$ Irreversible:

$(\Delta S)_{surr.} \neq -(\Delta S)_{sys.}$

$$= \frac{-q_{irr}}{T}\left(\text{Isothermal } T_{sys.} = T_{surr.}\right)$$

1st law:

$$\Delta U \overset{\nearrow 0}{=} q_{irr.} + W_{irr.}$$

$$\downarrow$$

$$-P_{ext}.\Delta U$$

2. In adiabatic ideal gas expansion/compression (dq = 0)

(i) If reversible adiabatic ($dq_{rev.} = 0$)

$$(\Delta S)_{sys.} = \int\frac{dq_{rev.}}{T} = 0$$

Reversible:

$(\Delta S)_{surr.} = -(\Delta S)_{sys.} = 0$

$\Rightarrow$ $(\Delta S)_{univ.} = 0$

(ii) If irreversible adiabatic ($dq_{irr.} = 0$)

$$(\Delta S)_{sys.} = \int\frac{(dq_{rev.})_{sys.}}{T} \neq 0$$

$$= n.C_{v,M.}\ln\frac{T_2}{T_1} + n.R.\ln\frac{V_2}{V_1}$$

For T_2 : Not PV^γ but 1st law

$$\Delta U = 0 + W_{irr.}$$

$$\downarrow \qquad \downarrow$$

$$n.C_{v,M}(T_2 - T_1) \qquad (-P_{ext}.\Delta V)$$

$\because$ **Irreversible:**

$$(\Delta S)_{Surr.} = \frac{\overset{\nearrow 0}{-q_{irr.}}}{T_{surr.}} = 0$$

So, $(\Delta S)_{univ.} = (\Delta S)_{Sys.}$

3. In the free expansion of ideal gas:

→ Always irreversible process.

→ Always isothermal as well as adiabatic.

$$\text{So, } (\Delta S)_{sys.} = \cancel{n.C_{v.M.} \ln \frac{T_2}{T_1}}^{0} + n.R.\ln \frac{V_2}{V_1}$$

$$= n.R.\ln \frac{V_2}{V_1} \text{ or } n.R.\ln \frac{P_1}{P_2} > 0$$

Irreversible:

$$(\Delta S)_{surr.} = \frac{-q_{irr.}}{T} = 0$$

$$\Rightarrow \quad (\Delta S)_{univ.} > 0$$

4. For chemical reaction/phase transition: (irreversible process)

$$(\Delta S)_{sys.} = \Delta_r S = \sum n_{(P)}.S_{M(P)} - \sum n_{(R)}.S_{M(R)}$$

$$\because \quad (\Delta S)_{surr.} = \frac{-q_{irr.}}{T}$$

reaction at constant volume:

$$(\Delta S)_{surr.} = \frac{-\Delta_r U}{T}$$

reaction at constant pressure:

$$(\Delta S)_{surr.} = \frac{-\Delta_r H}{T}$$

Q37 **Determine $(\Delta S)_{surr.}$ one mole of an ideal gas is expanded isothermally at 27°C until its volume is tripled if expansion is carried out reversibly.**

Sol:

$$(\Delta S)_{surr.} = -(\Delta S)_{sys.}$$

$$= \left[\cancel{n.C_{v,M} \ln \frac{T_2}{T_1}}^{0} + n.R.\ln \frac{V_2}{V_1} \right]$$

$$= -1.R.\ln \frac{3}{1}$$

$$= -R.\ln(3)$$

Q38 **The change in entropy of surroundings (ΔS_{surr}) in JK^{-1} is one mole of an ideal gas expands isothermally from 1.0 L to 2.0 L at 300 K in thermal contact with surroundings against a constant pressure of 3.0 atm. In this process,: (1Latm = 101.3J)**

(A) 6.763 **(B) 1.013** **(C) –1.013** **(D) –5.763**

Sol: **(C)**

$$(\Delta S)_{surr.} \neq -(\Delta S)_{sys.}$$

$$= \frac{-q_{irr.}}{T}$$

1st law:

$$\overset{0}{\cancel{dU}} = dq_{irr.} + dW_{irr}$$

$$W_{irr} = -P_{ext.}(V_2 - V_1)$$

$$= -3(2-1) = -303.9 \text{ J}$$

$$q_{irr.} = -w_{irr.} = +303.9 \text{ J}$$

Now: $(\Delta S)_{surr.} = \frac{-q_{irr.}}{T}$

$$= \frac{-303.9}{300} \text{J/K}$$

$$= -1.013 \text{ J/K}$$

Third law of thermodynamics:

At absolute zero temperature, the entropy of a perfect crystalline substance is zero.

Absolute zero temperature = Zero kelvin

So at zero kelvin, if the entropy of a substance is not zero, then the substance is not perfectly crystalline.

using this law, the absolute entropy of a perfect crystalline substance can be determined as follows:

Q39 **Determine entropy of $X_{(g)}$ at T kelvin.**

Sol: Let, $X_{(s)}$ (O.K.) $\longrightarrow$ $X_{(g)}$ (T.K.)

$$\Delta S = (S)_{final} - (S)^{\cancel{?}\,0}_{initial}$$

i.e., $\Delta S \Rightarrow$ entropy of $X_{(g)}$ at T kelvin

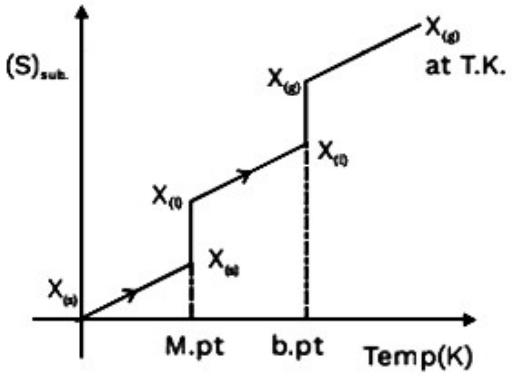
** Since crystalline substances have certain fixed values of their melting point and boiling point, means above process will take place as follows:

$$X_{(s)}\ (O.K.) \xrightarrow{\text{heating}} X_{(s)}\ (\text{melting point}) \xrightarrow{\text{melting}} X_{(l)}\ (\text{melting point}) \xrightarrow{\text{heating}} X_{(l)}\ (\text{boiling point}) \xrightarrow{\text{vaporization}} X_{(g)}\ (\text{boiling point}) \xrightarrow{\text{heating}} X_{(g)}\ (T.K.)$$

$$\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4 + \Delta S_5$$

$$\int_0^{M.pt} \frac{C_{(x,s)}.dT}{T} + \frac{\Delta H_{melting}}{M.pt.} + \int_{M.pt}^{b.pt} \frac{C_{(x,l)}.dT}{T} + \frac{\Delta H_{Vaporization}}{b.pt.} + \int_{b.pt}^{T} \frac{C_{(x,g)}.dT}{T}$$

Where, M.pt.- melting point
b.pt.- boiling point



Molar entropy (S_M):

For a substance,

$$S_M = \frac{\text{entropy}}{\text{moles}}\ J/Mol.K \text{ or } Cal/Mol.K$$

$\Rightarrow$ It is an intensive property.

Points to remember

for aqueous ions, S_M may be (+), (−), or even zero.

ΔG calculation in other processes:

for reversible:

$dU = dq_{rev.} + dw_{rev.}$

$dU = T.ds + (-P.dV)$

$\therefore \quad H = U + PV$

$dH = dU + P.dV + V.dP$

$dH = (TdS - PdV) + PdV + V.dP$

$dH = T.dS + V.dP$

$\therefore \quad G = H - TS$

$dG = dH - (T.dS + SdT)$

$dG = T.dS + VdP - T.dS - S.dT$

$\Rightarrow \quad dG = V.dP - S.dT$

Case I:

For pressure change at constant temperature:

$dT = 0$

$\Rightarrow \quad dG = V.dP$

For solids/liquid:

$V \rightarrow$ constant

$$\int dG = V.\int dP$$

$$\Rightarrow \quad (\Delta G)_T = V.(P_2 - P_1)$$

For gases:

$V = f(P)$

$$\Rightarrow \quad \Delta G = \int_{P_1}^{P_2} V.dP$$

For ideal gas:

$$\int dG = nRT\int \frac{1}{P}.dP$$

$$\Rightarrow \quad (\Delta G)_T = nRT \ln \frac{P_2}{P_1}$$

For reactions and phase transitions:

$d(\Delta_r G) = \Delta_r V.dP$

Let, at $P = P_1 \Rightarrow \Delta_r G = \Delta G_1$

at $P = P_2 \Rightarrow \Delta_r G = \Delta G_2$

Points to remember

As per IUPAC conversion

$S^{\circ}_{M(H^+,ac.)} = 0$

Standard molar entropy

$$\int_{\Delta G_1}^{\Delta G_2} d(\Delta_r G) = \Delta_r V.\int_{P_1}^{P_2} dP$$

$\Rightarrow \Delta G_2 - \Delta G_1 = \Delta_r V.(P_2 - P_1)$

Variation of $\Delta_r G$ with Pressure at a constant temperature

Where,

$\Delta_r V \rightarrow$ Change in volume for reaction

$= \Sigma(\text{Volume})_{\text{product}} - \Sigma(\text{Volume})_{\text{reactant}}$

Example:

$$X_{(S)} \longrightarrow X_{(l)}$$

$\Delta_r V = 1.V_{(X,l)} - 1.V_{(X,S)}$

Case II:

For temperature change at constant pressure:

$dP = 0$

$dG = -S.dT$

$\therefore$ $S = f_{(T)}$ always

So, for all substances:

$$(\Delta G)_p = -\int_{T_1}^{T_2} S.dT$$

For reactions and phase transition:

$\therefore$ $d(\Delta_r G) = -\Delta_r S.dT$

Let, at $T = T_1 \Rightarrow \Delta_r G = \Delta G_1$

at $T = T_2 \Rightarrow \Delta_r G = \Delta G_2$

$$\int_{\Delta G_1}^{\Delta G_2} d(\Delta_r G) = -\Delta_r S.\int_{T_1}^{T_2} dT$$

$\Delta G_2 - \Delta G_1 = -\Delta_r S(T_2 - T_1)$

Q40 **When 1.0 mole of H_2O (water) at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure, what is the free energy change (ΔG)?**

(A) 80 cal **(B) 540 cal** **(C) 620 cal** **(D) Zero**

Sol: **(D)**

100°C, 1 atm $\Rightarrow$ boiling point of water

$\Rightarrow$ liquid and gas in equilibrium

$\Rightarrow$ $\Delta G = 0$

Q41 **When 1.0 mole of H_2O (water) at 100°C and 1 atm pressure is converted into steam at 100°C and 2 atm pressure, what is the free energy change (ΔG)? [ln 2 = 0.693]**

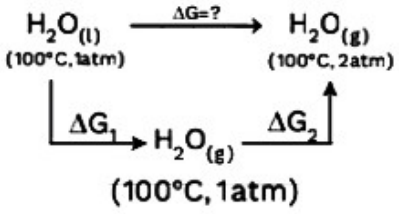
(A) zero cal **(B) 540 cal** **(C) 517.16 cal** **(D) 510 cal**

Sol: **(C)**

$\Delta G = \Delta G_1 + \Delta G_2$

$= O + nRT \ln \frac{P_2}{P_1}$

$= 1 \times 2 \times 373 \times \ln 2 = 517.13$ cal



Q42 **The standard state Gibbs free energies of formation of C(graphite) and C(diamond) at T = 298 K are** **[JEE Adv. 2017]**

$\Delta_f G°$ **[C(graphite)] = 0 kJ mol^{-1}**

$\Delta_f G°$ **[C(diamond)] = 2.9 kJ mol^{-1}**

The standard state means that the pressure should be 1 bar, and the substance should be pure at a given temperature. The conversion of graphite [C(graphite)] to diamond [C(diamond)] reduces its volume by 2×10^{-6} m^{3}mol^{-1}. If C (graphite) is converted to C(diamond) isothermally at T = 298 K, the pressure at which C(graphite) is in equilibrium with C(diamond) is :

[Useful information: 1 J = 1kg m^{2}s^{-2}; 1 Pa = 1 kgm^{-1}s^{-2}; 1 bar = 10^5 Pa]

(A) 14501 bar **(B) 29001 bar** **(C) 1450 bar** **(D) 58001 bar**

Sol: **(A)**

$C_{(S,graphite)} \longrightarrow C_{(S,diamond)}$

$\Delta_r G° = \Delta_f G°(diamond) - \Delta_f G°(graphite) = 2900$ J/mol

$= \Delta G$ at 1 bar (P_1)

Let, At P.bar, equilibrium $\Rightarrow$ DG = 0

$\Delta G_2 - \Delta G_2 = \Delta_r V (P_2 - P_1)$

$0 - 2900 = (-2 \times 10^{-6} M^3)(P - 1) \times 10^5$

J M^3 bar

$\frac{2900}{2 \times 10^{-6} \times 10^5} = P - 1$

$P - 1 = 14500$

$P = 14501$ bar

Thermodynamic analysis of chemical equilibrium:

for a reversible reaction taking place at constant temperature and pressure, at any time,

$\Delta G = \Delta G^\circ + RT \ln Q^\circ$

Where, ΔG = free energy of reaction at nonstandard state (J/mol) or (cal/mol)

ΔG° = free energy of reaction at standard state (J/mol) or (cal/mol)

R = 8.314 J/mol.K = 2 cal/mol.K

T = kelvin temperature

Q° = Thermodynamic reaction quotient (unitless)

Thermodynamic analysis of chemical equilibrium:

$\Delta G = \Delta G^\circ + RT \ln Q^\circ$

$Q^\circ \rightarrow$ Thermodynamic reaction quotient (unitless)

For a reversible reaction:

$$Q^\circ = \frac{\pi(a)^{\upsilon_{product}}_{product}}{\pi(a)^{\upsilon_{reactant}}_{reactant}}$$

$\upsilon \rightarrow$ Stoichiometric coefficient

$q \rightarrow$ activity (unitless)

For solids/liquids

$a = 1$

For aqueous

$$a = \left(\frac{[\,]}{1M}\right)$$

For gas

$$a = \frac{P_{gas}(1bar)}{1bar}$$

E.g. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$$Q^\circ = \frac{(P_{NO_2} / 1bar)^2}{(P_{N_2O_4} / 1bar)^1} unitless$$

$$Q = \frac{(P_{NO_2})^2}{(P_{N_2O_4})} bar$$

But, $|Q^\circ| = |Q|$

E.g., $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$$Q = (P_{CO_2})^1$$

E.g., $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$Q = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^3}$$

E.g., $Zn(s) + 2Ag^+_{(aq.)} \rightleftharpoons Zn^{2+}_{(aq.)} + 2Ag(s)$

$$Q = \frac{[Zn^{2+}_{(aq.)}].1}{1.[Ag^+_{(aq.)}]^2}$$

E.g., $Fe(s) + 2H^+_{(aq.)} \rightleftharpoons Fe^{2+}_{(aq.)} + H_2(g)$

$$Q = \frac{[Fe^{2+}_{(aq)}]^1 \left(P_{H_2}\right)^1}{1.[H^+_{(aq)}]^2}$$

In the above expressions, if the activity of equilibrium time is put, then K°_{eq} (unitless) will be obtained.

Now again:

$\because \quad \Delta G = \Delta G^\circ + RT \ln Q^\circ$

At equilibrium:

$Q^\circ \Rightarrow K^\circ_{eq}$ and $\Delta G = 0$

So, $0 = \Delta G^\circ + Rt \ln K^\circ_{eq}$

$\Rightarrow \quad \Delta G^\circ = -RT. \ln K^\circ_{ec}$

$= -2.303\ RT \log K^\circ_{eq}$

Since the value of K_{eq} gives information about the extent of reaction so now ΔG° can also give information of extant as following:

Case I:

If $K_{eq} >> 1 \Rightarrow \Delta G^\circ << 0 \Rightarrow$ high conversion of R into P

Also, $\because \Delta G^\circ = (G^\circ)_p - (G^\circ)_R$

i.e., $G^\circ_R >> G^\circ_P$

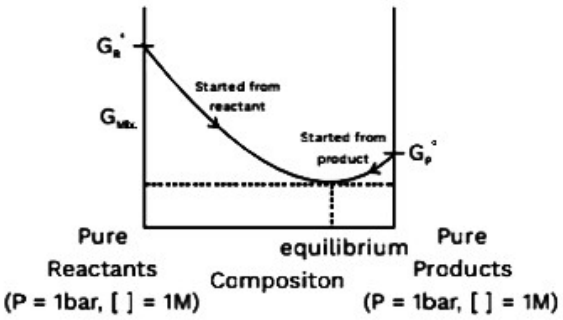
Case II:

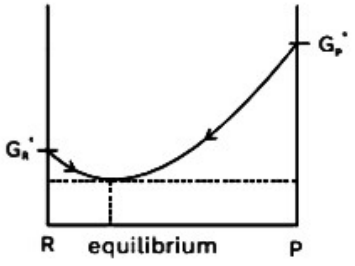
If $K_{eq} << 1 \Rightarrow \Delta G^\circ >> 0 \Rightarrow$ low conversion of R into P

Also, $\because \Delta G^\circ = (G^\circ)_p - (G^\circ)_R$

$\Rightarrow \quad (G^\circ)_p >> (G^\circ)_R$

For $R \rightleftharpoons P$



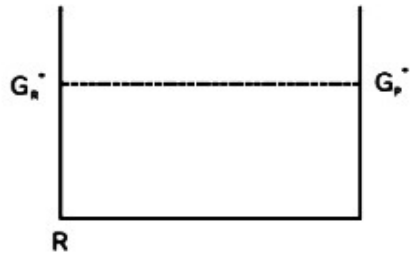


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Case III:

If $K_{eq} \sim 1 \Rightarrow \Delta G^\circ \sim 0 \Rightarrow$ Moderate conversion

Also, $(G^\circ)_P \sim (G^\circ)_R$



Case IV:

If $K_{eq} \sim 1$

Also, For a reaction :

$\because \quad \Delta G^\circ = \Delta H^\circ - T.\Delta S^\circ$

$-RT \ln K_{eq} = \Delta H^\circ - T.\Delta S^\circ$

$$\Rightarrow \quad \ln K_{eq} = = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

$$\text{Also, } \log K_{eq} = \frac{-\Delta H^\circ}{2.303RT} + \frac{\Delta S^\circ}{2.303R}$$

For $\ln K_{eq}$ v/s $\frac{1}{T}$ graph

$$\because \quad \underset{\downarrow y}{\ln K_{eq.}} = -\frac{\Delta H^\circ}{R}\left(\underset{\downarrow x}{\frac{1}{T}}\right) + \frac{\Delta S^\circ}{R}$$

$\Rightarrow \quad y = Mx + c$

Where,

$$\text{Slope} = \left(\frac{-\Delta H^\circ}{R}\right)$$

> 0 for exothermic

< 0 for endothermic

$$\text{Intercept} = \left(\frac{\Delta S^\circ}{R}\right)$$

If, $\Delta n_g > 0 \Rightarrow \Delta S^\circ > 0$

$\Delta n_g < 0 \Rightarrow \Delta S^\circ < 0$

E.g., $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

$\because$ Dissociation $\Rightarrow \Delta H^\circ > 0 \Rightarrow M < 0$

Also, $\Delta n_g > 0 \Rightarrow \Delta S^\circ > 0 \Rightarrow C > 0$

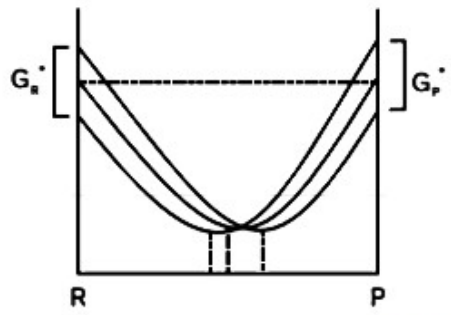
Effect of temperature on a reversible reaction:

Consider ΔH° and ΔS° to be temperature independent.

$\Rightarrow \quad K_{eq} = f_{(T)}$

Let, $T = T_1$

$$\ln K_1 = \frac{-\Delta H^\circ}{RT_1} + \frac{\Delta S^\circ}{R}$$



Points to remember

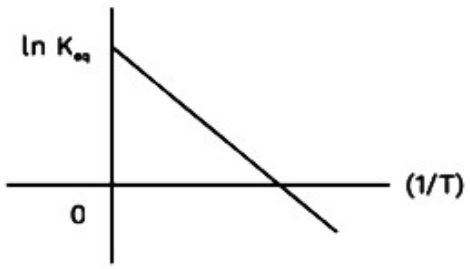
$\Rightarrow$ Hence the gibers free energy is minimum at equilibrium.

$\Rightarrow$ Since at any time, Q gives information about the direction for equilibrium so now ΔG can also be used for direction.

If $\Delta G < 0 \Rightarrow Q < K_{eq} \Rightarrow$ forward

$\Delta G > 0 \Rightarrow Q > K_{eq} \Rightarrow$ backward

$\Delta G > 0 \Rightarrow Q > K_{eq} \Rightarrow$ equilibrium

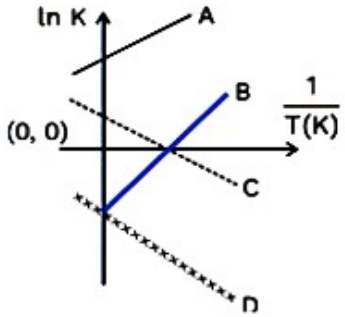


Let, $T = T_2$

$$\ln K_2 = \frac{-\Delta H^\circ}{RT_2} + \frac{\Delta S^\circ}{R}$$

$$\Rightarrow \quad \boxed{\ln \frac{K_2}{K_1} = \frac{\Delta H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

43 **Select the correctly lines show the temperature dependence of equilibrium constant, K, for an exothermic reaction?** **[JEE Main 2018]**



(A) C and D **(B) A and D** **(C) A and B** **(D) B and C**

Sol: **(C)**

$\Delta H^\circ < 0$

$$\Rightarrow \text{ Slope} = \left(\frac{-\Delta H^\circ}{R}\right) > 0$$

44 **Consider the following reversible reaction,**

$A(g) + B(g) = AB(g)$

The activation energy of the backward reaction exceeds that of the forward reaction by 2RT (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 4 times that of the reverse reaction, the absolute value of ΔG° (in J mol^{-1}) for the reaction at 300 K is ____.

(Given; ln(2) = 0.7, RT = 2500 J mol^{-1} at 300 K and G is the Gibbs energy)

Sol: $E_{ab} = E_{af} + 2RT$

$\because \quad \Delta H^\circ = E_{af} - E_{ab}$

$= (-2RT)$

$$K_{eq} = \frac{K_f}{K_b} = \frac{A_f.e^{-E_{af}/RT}}{A_b.e^{-E_{ab}/RT}}$$

$$K_{eq} = \frac{4}{1}e^{-\left(\frac{-2RT}{RT}\right)} = 4.e^2$$

$\because \quad \Delta G^\circ = -RT\ln(4.e^2)$

$= -RT[2\ln e + \ln 4]$

$= -2500[2 + 1.4]$

$= -8500$ J/mol